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System and method for using intelligent staggered telemetry for node capabilities management across nodes in a peripheral device workspace

Abstract

A system and method for a peripheral device workspace cloud orchestrator server to receive environmental telemetry data associated with a plurality of device nodes within a peripheral device workspace at a location that includes an anchor information handling system node with a hardware processor to execute computer-readable program code of a telemetry ownership orchestrator module to determine to offload a portion of environmental telemetry data collection and aggregation functions to be transitioned to a smart device node within the peripheral device workspace from the anchor information handling system node when a resource utilization level at the anchor information handling system node is determined to reach a utilization threshold, and transmitting instructions to the anchor information handling system node to offload the portion of environmental telemetry data collection and aggregation functions for a subset of the plurality of peripheral device nodes in the peripheral device workspace to the smart device node.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to automatic orchestration and collection of telemetry data of peripheral device workspace nodes within a peripheral device workspace. The present disclosure more specifically relates to orchestrating and collecting telemetry data from information

handling system nodes, smart device nodes, and peripheral device nodes within a peripheral device workspace to manage various capabilities of those nodes related to the set of detected operating context data of executing applications of anchor information handling system nodes such as a user's information handling system or smart nodes within that peripheral device workspace.

BACKGROUND

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to clients is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing clients to take advantage of the value of the information. Because technology and information handling may vary between different clients or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific client or specific use, such as e-commerce, financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems. The information handling system may include telecommunication, network communication, and video communication capabilities. The information handling system may be used to execute instructions of one or more applications such as workplace productivity applications or a gaming application. The information handling system may be operatively coupled to a cloud server information handling system via a network connection. Further, the information handling system may be operatively coupled to a plurality of peripheral devices that include smart peripheral devices at a location and the information handling system may be operatively coupled to a cloud server information handling system via a network connection.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:

(2) FIG. 1 is a block diagram illustrating an information handling system operatively coupled to a remotely located peripheral device workspace cloud orchestrator server for management of peripheral device workspaces in an enterprise according to an embodiment of the present disclosure;

(3) FIG. 2 is a graphic diagram of an anchor information handling system node forming a peripheral device workspace and including a plurality of peripheral device nodes that is operatively coupled to a peripheral device workspace cloud orchestrator server according to an embodiment of the present disclosure;

(4) FIG. 3 is a flow chart showing a method of executing code instructions of a cloud manageability orchestrator module on a peripheral device workspace cloud orchestrator server to automatically register node devices within one or more peripheral device workspaces to manage and orchestrate those node devices according to an embodiment of the present disclosure;

(5) FIG. 4 is a flow chart showing a system and method of computing telemetry offloading within a peripheral device workspace according to an embodiment of the present disclosure according to an

embodiment of the present disclosure;

(6) FIG. 5 is a flow chart showing a system and method of generating capability mapping recommendations within a peripheral device workspace according to an embodiment of the present disclosure; and

(7) FIG. 6 is a flow chart showing method of staggering telemetry for node capabilities management across device nodes in a peripheral device workspace according to an embodiment of the present disclosure.

(8) The use of the same reference symbols in different drawings may indicate similar or identical items.

DETAILED DESCRIPTION OF THE DRAWINGS

(9) The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.

(10) A user may oftentimes use an information handling system at a physical location in a peripheral device workspace. In this context, a peripheral device workspace can be viewed as an environment that includes a user information handling system (e.g., a laptop) as an anchor information handling system node or primary node and one or more peripheral devices as peripheral device nodes also referred to as peripherals that are connected to the user information handling system at an identified location. Any number of peripheral device may be operatively coupled to the anchor information handling system node and may include smart device nodes that include, in some example embodiments, a hardware processing resource and/or network communication capabilities. The location, and a manifest of these device nodes at that physical location define a peripheral device workspace that is associated with a peripheral device workspace identification value. In an embodiment, some of the peripheral devices operatively coupled to the anchor information handling system node may be a smart peripheral device. A smart peripheral device may include any peripheral device that can be operatively coupled to a network as well as those peripheral devices that include computing resources such as data storage resources, processing resources, and power resources, among other resources. In an embodiment, each of the information handling systems, smart peripheral devices, and peripheral devices within these peripheral device workspaces may be referred to as device nodes and form part of these peripheral device workspaces.

(11) According to embodiments herein, a formed peripheral device workspace may oftentimes be used for various work scenarios. For example, a business may have an office space that includes hoteling cubes that can be assigned to, reserved by, or otherwise utilized by the business' employees as peripheral device workspaces for use with one or more peripheral devices and an information handling system introduced to the peripheral device workspace for the duration it is used by the identified user, for example, a business' employee. In such an example case, the business may allow its individual employees to connect their laptops to a docking station or to peripheral devices directly in a particular hoteling cube having a formed peripheral device workspace where various external peripherals may be available for use. The docking station, in some example embodiments, may be a smart peripheral device that includes a hardware processing device, a data storage device, and/or a wireless radio device capable of operatively coupling the docking station to a network in some example embodiments. Users may also employ other peripheral device workspaces when working from home or other locations and the information handling system and some portion of the external peripheral devices may travel with the user to one or more of the identified peripheral device workspaces that a particular user may enter and use. The plurality of peripheral device workspaces associated with a user and a user identification may be defined under a composite user peripheral device workspace identifier and stored in a cloud orchestrator database.

(12) As users are interacting with the peripheral device nodes including the anchor information handling system nodes, smart peripheral devices, and other peripheral device nodes, the anchor information handling system node may collect telemetry data. This telemetry data may include, for example, any data, in situ of each of the peripheral device nodes. Some specific examples of telemetry data may include, but are not limited to, Input/Output (I/O) operation load on the peripheral device nodes, memory usage the peripheral device nodes, storage usage the peripheral device nodes, CPU/hardware processor power utilization of the peripheral device nodes, connectivity types, connectivity quality, application and/or hardware uptime and downtime of the peripheral device nodes, user information, application errors and exceptions, database accesses of each of the peripheral device nodes, latency, web log information, and database errors and warnings. This telemetry data may be generated at and received from each of the individual peripheral device nodes in a peripheral device workspace. However, this telemetry data is typically collected at an anchor information handling system node such as an anchor information handling system node. Typically, this telemetry data is then used by the information handling system to provide orchestration recommendations for the peripheral device nodes across the peripheral device workspace. However, with the anchor information handling system node executing software, firmware, and application data via its hardware processor, the collection of the telemetry data and calculations related to the provision of recommendations for management of a peripheral device workspace may consume valuable processing resources reducing the quality of experience (QoE) by the user at the anchor information handling system node.

(13) The present specification describes computer-readable program code of an ecosystem manageability service module executing at a peripheral device workspace cloud orchestrator server that, when executed by a hardware processor of the peripheral device workspace cloud orchestrator server, directs for telemetry data to be collected by a plurality of peripheral device workspace telemetry sub-agents executed by the hardware processors of each anchor information handling system nodes. In an embodiment, execution of computer-readable program code of a telemetry aggregator module selectively aggregates this telemetry data from an anchor information handling system node and/or smart device node depending on key performance indicator (KPI) data received from each of the device nodes within the peripheral device workspace. Additionally, each of the anchor information handling system node and one or more smart device nodes may execute computer-readable program code of an ecosystem manageability sub-agent that sends the KPI data related to each peripheral device nodes, smart device nodes, and the anchor information handling system node to the peripheral device workspace cloud orchestrator server executing the computer-readable program code of the telemetry aggregator module. This KPI data may be included within the telemetry data and may include processing performance KPI, battery availability KPI, audio KPIs, video KPIs, and the like that is used to orchestrate the offloading of collection and assessment of telemetry data collected by the anchor information handling system node and smart device nodes when KPI threshold levels indicate that one of the anchor information handling system node or smart device nodes should have the telemetry gathering responsibilities associated with their respective peripheral device nodes or accessing offloaded to it. This may be done by the peripheral device workspace cloud orchestrator server directing, for example, the smart device node to conduct the telemetry orchestration (e.g., collection and assessment of telemetry data) from each device node within the peripheral device workspace based on currently-received peripheral device telemetry data related to the KPIs and received at the peripheral device workspace cloud orchestrator server. For example, the peripheral device workspace may include the anchor information handling system node that is operatively coupled directly to secondary peripheral device nodes of a keyboard and mouse as well as to a docking station. The docking station, in turn, may be operatively coupled to secondary peripheral device nodes of an external monitor, a webcam, and an audio device such as a speaker or a headset. Typically, the anchor information handling system node may handle all the telemetry collection and aggregation responsibilities even

for the external monitor, webcam and audio device operatively coupled to the docking station. However, the hierarchy of this telemetry data collection may be changed from the anchor information handling system node being solely responsible for this collection and aggregation to the anchor information handling system node and docking station splitting responsibility for the collection and aggregation of telemetry data associated with their owned, respective peripheral device nodes. Each of the anchor information handling system node and smart node device which may split collection and aggregation of telemetry data may have network access to the peripheral device workspace cloud orchestration server managing the peripheral device workspace. The peripheral device workspace cloud orchestrator server, in this example embodiment, may determine when any offloading of telemetry collection and aggregation as it relates to individual peripheral device nodes within the workspace is to occur due to high level of processing, memory, or communications activity at the anchor information handling system node or the smart device node.

(14) In an embodiment, the peripheral device workspace cloud orchestrator server may execute the computer-readable program code of a telemetry ownership orchestrator module to map the telemetry data received and aggregated from the anchor information handling system node to each device node in the peripheral device workspace and determine, based on the KPI data and device node capabilities, whether telemetry aggregation functions are to be, at least partially, transitioned to a smart device node within the peripheral device workspace. The execution of the telemetry ownership orchestrator module to send a transitional offload mapping to the anchor information handling system node and smart device nodes that indicates, based on current KPI data, which of the anchor information handling system node and smart device node is to conduct this telemetry orchestration. This transitional offload mapping may indicate to either of the chosen anchor information handling system node or smart device node those peripheral devices that are to send telemetry data and, based on the received telemetry data, orchestrate operational recommendations for the peripheral device nodes across the peripheral device workspace.

(15) As described herein, the environmental telemetry data for nodes in a peripheral device workspace may include any data that may affect the QoE by a user of the peripheral device workspace. For example, this environmental telemetry data may be data descriptive of current and available processing resources in each peripheral device node (including the anchor information handling system node and smart device node), current and available resources in each peripheral device node, current power sources of each of the peripheral device nodes, current battery levels within each peripheral device nodes, current methods of operatively coupling the smart nodes and peripheral devices to the anchor information handling system node, as well as executing software applications and software data (e.g., scheduled videoconferencing sessions and length of the videoconferencing sessions), among other environmental telemetry data.

(16) When the ecosystem manageability sub-agent of the selected anchor information handling system node or smart device node receives this environmental telemetry data from each of the peripheral device nodes being monitored within the peripheral device workspace, it may cache this environmental telemetry data at, for example, a telemetry database in a cloud orchestrator database for use during management of operation of nodes within the peripheral device workspace. Further, execution of the computer-readable program code of the ecosystem manageability sub-agent causes the ecosystem manageability sub-agent to analyze trends associated with the use of the anchor information handling system node, smart nodes, and peripheral devices as well as the execution of applications on the anchor information handling system nodes to determine whether QoE notifications should be sent out to the anchor information handling system nodes and users operating the anchor information handling system nodes indicating that changes to the QoE will be experienced unless changes to operations of the anchor information handling system node, smart peripheral device nodes, and peripheral devices are taken by the user.

(17) In an embodiment, the execution of computer-readable program code of a workspace capability orchestrator module by the cloud orchestrator hardware processing device of the

peripheral device workspace cloud orchestrator server allows for the peripheral device workspace cloud orchestrator server to identify duplicate or overlapping capabilities among peripheral device nodes within any created peripheral device workspace. Additionally, this duplicate or overlapping capability data among the peripheral device nodes may be accessed and used to select between capabilities at those nodes or prioritization among capabilities at those peripheral device nodes to maintain a QoE. For example, where the peripheral device workspace includes a web cam that is built into the chassis of the anchor information handling system node and an external web cam is also operatively coupled to the anchor information handling system node, the selection of which web cam to use may be based on which web cam has the best image sensor, resolution, largest number of potential settings available, and other characteristics of the two web cams. This may increase the QoE for the user.

(18) The systems and methods described herein, therefore, allows the anchor information handling system node and operatively-coupled smart device nodes to cooperate in the collection of environmental telemetry data for transmission to the peripheral device workspace cloud orchestrator server which may execute code instructions of a peripheral device workspace cloud orchestrator for orchestration of operational recommendations to each peripheral device node within the peripheral device workspace. This allows the processing resources within, for example, the anchor information handling system node to be used for the execution of other applications thereby reducing the consumption of processing resources at the anchor information handling system node. This increases the QoE realized at the anchor information handling system node even when processing-intensive applications are being executed on the anchor information handling system node in a peripheral device workspace.

(19) Turning now to the figures, FIG. 1 illustrates an anchor information handling system node **100** similar to the information handling systems according to several aspects of the present disclosure. In the embodiments described herein, an anchor information handling system node **100** includes any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or use any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, an anchor information handling system node **100** may be a personal computer, mobile device (e.g., personal digital assistant (PDA) or smart phone), server (e.g., blade server or rack server), a consumer electronic device, a network server or storage device, a network router, switch, or bridge, wireless router, or other network communication device, a network connected device (cellular telephone, tablet device, etc.), IoT computing device, wearable computing device, a set-top box (STB), a mobile information handling system, a palmtop computer, a laptop computer, a desktop computer, a communications device, an access point (AP) **138**, a base station transceiver **140**, a wireless telephone, a control system, a camera, a scanner, a printer, a personal trusted device, a web appliance, or any other suitable machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine, and may vary in size, shape, performance, price, and functionality. It is appreciated that, in some embodiments herein, the information handling system **100** may be an anchor information handling system node (also referred to herein as the anchor information handling system node **100**) and may be one of a plurality of device smart nodes or user information handling system nodes as part of a peripheral device workspace. Systems similar to the information handling system **100** may also operate as one or more peripheral device workspace cloud orchestrator servers **158** that are operatively coupled to a peripheral device workspace cloud orchestrator consol **160** graphical user interface and may execute code instructions of one or more software modules or systems of the peripheral device workspace cloud orchestrator **156** described herein. In an embodiment, the cloud orchestrator consol **160** may be an information handling system **100** itself that is used by an internet technology decision maker (ITDM) to create hardware device operational policies with one or more peripheral device workspace cloud orchestrator servers **158** to be propagated down to

device nodes within a collaborative peripheral device workspace such as the anchor information handling system node **100**, a docking station **151**, video display device **144**, keyboard **146**, stylus **148**, trackpad **150**, mouse **152**, wireless headset **153**, and the like which may be attached or separate devices from anchor information handling system node **100** in embodiments described herein. In this embodiment, the anchor information handling system node **100**, may receive the hardware device operational policies generated by the ITDM at the peripheral device workspace cloud orchestrator consol **160** via execution of code instructions of the cloud manageability orchestrator module **168** at the peripheral device workspace cloud orchestrator server **158** as described in embodiments herein.

(20) Thus, in a networked deployment, the information handling system **100** may operate in the capacity of a client computer in a server-client network environment, or as a peer computer system in a peer-to-peer (or distributed) network environment. In an embodiment, the information handling system node **100** may be implemented using electronic devices that provide voice, video, or data communication. For example, an information handling system **100** may be any mobile or other computing device capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single anchor information handling system node **100** is illustrated, the term “system” shall also be taken to include any collection of systems or sub-systems that individually or jointly execute a set, or plural sets, of instructions to perform one or more computer functions.

(21) The information handling system **100** may include main memory **106**, (volatile (e.g., random-access memory, etc.), or static memory **108**, nonvolatile (read-only memory, flash memory etc.) or any combination thereof), one or more hardware processing resources, such as a hardware processor **102** that may be a central processing unit (CPU), a graphics processing unit (GPU) **103**, embedded controller (EC) **104**, or any combination thereof. Additional components of the information handling system **100** may include one or more storage devices such as static memory **108** or drive unit **120**. The information handling system **100** may include or interface with one or more communications ports for communicating with external devices, as well as various input and output (I/O) devices **142**, such as a docking station **151**, a mouse **152**, a trackpad **150**, a keyboard **146**, a stylus **148**, a video/graphics display device **144**, or any combination thereof which may be internal and/or external devices to the anchor information handling system node **100** in various embodiments. Portions of an information handling system **100** may themselves be considered information handling systems **100**.

(22) Information handling system **100** may include devices or modules that embody one or more of the devices or execute instructions for one or more systems and modules. The information handling system **100** may execute instructions (e.g., software algorithms), parameters, and profiles **112** that may operate on servers or systems, remote data centers, or on-box in individual client information handling systems according to various embodiments herein. In some embodiments, it is understood any or all portions of code instructions (e.g., software algorithms), parameters, and profiles **112** may operate on a plurality of information handling systems **100**. Example software or firmware sets of executable code instructions may include the peripheral device workspace cloud orchestrator **156** which may include plural software modules such as cloud manageability orchestrator module **168**, ecosystem manageability service module **170**, cloud orchestrator notification module **172**, telemetry ownership orchestrator module **174**, telemetry aggregator module **176**, workspace capability orchestrator module **178**, and workspace context discovery module **180**, among others such as operating systems and the like on an information handling system **100** operating as a peripheral device workspace cloud orchestrator server **158** in some embodiments. Other example software or firmware sets of executable code instructions may include the telemetry aggregator sub-agent **184-1**, **184-2** and ecosystem manageability and telemetry sub-agent **186-1**, **186-2** among others such as operating systems **116** and the like on an information handling system **100** operating as an anchor information handling system node at a

peripheral device workspace in some embodiments.

(23) The information handling system **100** may include the hardware processor **102** such as a central processing unit (CPU). Any of the processing resources may operate to execute code that is either firmware or software code. Moreover, the information handling system **100** may include memory such as main memory **106**, static memory **108**, and disk drive unit **120** (volatile (e.g., random-access memory, etc.), nonvolatile memory (read-only memory, flash memory etc.) or any combination thereof or other memory with computer readable medium **110** storing instructions (e.g., software algorithms), parameters, and profiles **112** executable by the EC **104**, hardware processor **102**, GPU **103**, or any other hardware processing device. The anchor information handling system node **100** may also include one or more buses **118** operable to transmit communications between the various hardware components and peripheral devices such as any combination of various I/O devices **142** as well as between hardware processors **102**, an EC **104**, the operating system (OS) **116**, the basic input/output system (BIOS) **114**, the wireless interface adapter **128**, or a radio module, among other components described herein. In an embodiment, the anchor information handling system node **100** may be in wired or wireless communication with the I/O devices **142** such as a docking station **151**, a keyboard **146**, a mouse **152**, video display device **144**, stylus **148**, or trackpad **150** among other peripheral devices that may be peripheral device nodes in a peripheral device workspace.

(24) The information handling system **100** further includes a video/graphics display device **144**. The video/graphics display device **144** in an embodiment may function as a liquid crystal display (LCD), an organic light emitting diode (OLED), a flat panel display, or a solid-state display. It is appreciated that the video/graphics display device **144** may be wired or wireless and may be an external video/graphics display device **144** that allows a user to increase the desktop area by extending the desktop in an embodiment.

(25) Additionally, as described herein, the information handling system **100** serving as an anchor information handling system node **100** may include or be operatively coupled to one or more other I/O devices **142** including the wired or wireless mouse **152** described herein that allows the user to interface with the anchor information handling system node **100** via the video/graphics display device **144**, a cursor control device (e.g., a trackpad **150**, or gesture or touch screen input), a stylus **148**, and/or a keyboard **146**, among others. The anchor information handling system node **100** may also be operatively coupled to any peripheral device **142** such as a docking station **151** or other smart peripheral device having a hardware processing device such as a hardware processor, microcontroller, or other hardware processing resource and which may further be operatively coupled to one or more additional peripheral devices **142**. As described herein, each of these input/output devices **142** may be a device node associated with the anchor information handling system node **100** and may be part of a peripheral device workspace on its own and identified with a peripheral device workspace identification value via execution of the cloud manageability orchestrator module **168** as described herein. Various drivers and hardware control device electronics may be operatively coupled to operate the I/O devices **142** according to the embodiments described herein. The present specification contemplates that the I/O devices **142** may be wired or wireless.

(26) A network interface device of the information handling system **100**, such as wireless interface adapter **128** or a wired network interface device can provide connectivity among devices such as with Bluetooth® or to a network **136**, e.g., a wide area network (WAN), a local area network (LAN), wireless local area network (WLAN), a wireless personal area network (WPAN), a wireless wide area network (WWAN), or other network. In an embodiment, this network **136** may be operatively coupled to or include a peripheral device workspace cloud orchestrator **156** that includes one or more servers (e.g., peripheral device workspace cloud orchestrator server **158**) or other computing devices that provide computer system resources as described herein that allow for the creation of peripheral device workspaces and orchestration of different device nodes within one

or more peripheral device workspaces. In embodiments described herein, the wireless interface device **128** with its radio **130**, RF front end **132** and antenna **134** is used to communicate with the wireless peripheral devices via, for example, a Bluetooth® or Bluetooth® Low Energy (BLE) protocols. In an embodiment, the WAN, WWAN, LAN, and WLAN may each include an AP **138** or base station **140** used to operatively couple the anchor information handling system node **100** to a network **136**. In a specific embodiment, the network **136** may include macro-cellular connections via one or more base stations **140** or a wireless AP **138** (e.g., Wi-Fi), or such as through licensed or unlicensed WWAN small cell base stations **140**. Connectivity may be via wired or wireless connection. For example, wireless network wireless APs **138** or base stations **140** may be operatively connected to the anchor information handling system node **100**. Wireless interface adapter **128** may include one or more radio frequency (RF) subsystems (e.g., radio **130**) with transmitter/receiver circuitry, modem circuitry, one or more antenna radio frequency (RF) front end circuits **132**, one or more wireless controller circuits, amplifiers, antennas **134** and other circuitry of the radio **130** such as one or more antenna ports used for wireless communications via multiple radio access technologies (RATs). The radio **130** may communicate with one or more wireless technology protocols.

(27) In an embodiment, the wireless interface adapter **128** may operate in accordance with any wireless data communication standards. To communicate with a wireless local area network, standards including IEEE 802.11 WLAN standards (e.g., IEEE 802.11ax-2021 (Wi-Fi 6E, 6 GHz)), IEEE 802.15 WPAN standards, WWAN such as 3GPP or 3GPP2, Bluetooth® standards, or similar wireless standards may be used. Wireless interface adapter **128** may connect to any combination of macro-cellular wireless connections including 2G, 2.5G, 3G, 4G, 5G or the like from one or more service providers. Utilization of radio frequency communication bands according to several example embodiments of the present disclosure may include bands used with the WLAN standards and WWAN carriers which may operate in both licensed and unlicensed spectrums. The wireless interface adapter **128** can represent an add-in card, wireless network interface module that is integrated with a main board of the anchor information handling system node **100** or integrated with another wireless network interface capability, or any combination thereof.

(28) In some embodiments, software, firmware, dedicated hardware implementations such as application specific integrated circuits, programmable logic arrays and other hardware devices may be constructed to implement one or more of some systems and methods described herein. Applications that may include the apparatus and systems of various embodiments may broadly include a variety of electronic and computer systems. One or more embodiments described herein may implement functions using two or more specific interconnected hardware modules or devices with related control and data signals that may be communicated between and through the modules, or as portions of an application-specific integrated circuit. Accordingly, the present system encompasses software, firmware, and hardware implementations.

(29) In accordance with various embodiments of the present disclosure, the methods described herein may be implemented by firmware or software programs executable by a hardware controller or a hardware processor system. Further, in an exemplary, non-limited embodiment, implementations may include distributed hardware processing, component/object distributed hardware processing, and parallel hardware processing. Alternatively, virtual computer system processing may be constructed to implement one or more of the methods or functionalities as described herein.

(30) The present disclosure contemplates a computer-readable medium that includes instructions, parameters, and profiles **112** or receives and executes instructions, parameters, and profiles **112** responsive to a propagated signal, so that a hardware device connected to a network **136** may communicate voice, video, or data over the network **136**. Further, the instructions **112** may be transmitted or received over the network **136** via the network interface device or wireless interface adapter **128**. It is appreciated that any computing device including the cloud orchestrator server

158, the cloud orchestrator console **160**, and the anchor information handling system node **100** may include a computer-readable medium that includes instructions, parameters, and profiles **112**.

(31) The information handling system **100** may include a set of instructions **112** that may be executed to cause the computer system to perform any one or more of the methods or computer-based functions disclosed herein. For example, instructions **112** may be executed by a hardware processor **102**, GPU **103**, EC **104** or any other hardware processing resource and may include software agents, or other aspects or components used to execute the methods and systems described herein. Various software modules comprising application instructions **112** may be coordinated by an OS **116**, and/or via an application programming interface (API). An example OS **116** may include Windows®, Android®, and other OS types. Example APIs may include Win 32, Core Java API, or Android APIs.

(32) In an embodiment, the information handling system **100** may include a disk drive unit **120**. The disk drive unit **120** and may include machine-readable code instructions, parameters, and profiles **112** in which one or more sets of machine-readable code instructions, parameters, and profiles **112** such as firmware or software can be embedded to be executed by the hardware processor **102** or other hardware processing devices such as a GPU **103** or EC **104**, or other microcontroller unit to perform the processes described herein. Similarly, main memory **106** and static memory **108** may also contain a computer-readable medium for storage of one or more sets of machine-readable code instructions, parameters, or profiles **112** described herein. The disk drive unit **120** or static memory **108** also contain space for data storage. Further, the machine-readable code instructions, parameters, and profiles **112** may embody one or more of the methods as described herein. In a particular embodiment, the machine-readable code instructions, parameters, and profiles **112** may reside completely, or at least partially, within the main memory **106**, the static memory **108**, and/or within the disk drive **120** during execution by the hardware processor **102**, EC **104**, or GPU **103** of anchor information handling system node **100**.

(33) Main memory **106** or other memory of the embodiments described herein may contain computer-readable medium (not shown), such as RAM in an example embodiment. An example of main memory **106** includes random access memory (RAM) such as static RAM (SRAM), dynamic RAM (DRAM), non-volatile RAM (NV-RAM), or the like, read only memory (ROM), another type of memory, or a combination thereof. Static memory **108** may contain computer-readable medium (not shown), such as NOR or NAND flash memory in some example embodiments. The applications and associated APIs, for example, may be stored in static memory **108** or on the disk drive unit **120** that may include access to a machine-readable code instructions, parameters, and profiles **112** such as a magnetic disk or flash memory in an example embodiment. While the computer-readable medium is shown to be a single medium, the term “computer-readable medium” includes a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of machine-readable code instructions. The term “computer-readable medium” shall also include any medium that is capable of storing, encoding, or carrying a set of machine-readable code instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein.

(34) In an embodiment, the information handling system **100** may further include a power management unit (PMU) **122** (a.k.a. a power supply unit (PSU)). The PMU **122** may include a hardware controller and executable machine-readable code instructions to manage the power provided to the components of the information handling system **100** such as the hardware processor **102** and other hardware components described herein. The PMU **122** may control power to one or more components including the one or more drive units **120**, the hardware processor **102** (e.g., CPU), the EC **104**, the GPU **103**, a video/graphic display device **144**, or other wired I/O devices **142** such as the mouse **152**, the stylus **148**, a keyboard **146**, and a trackpad **150** and other components that may require power when a power button has been actuated by a user. In an

embodiment, the PMU **122** may monitor power levels and be electrically coupled to the information handling system **100** to provide this power. The PMU **122** may be coupled to the bus **118** to provide or receive data or machine-readable code instructions. The PMU **122** may regulate power from a power source such as the battery **124** or AC power adapter **126**. In an embodiment, the battery **124** may be charged via the AC power adapter **126** and provide power to the components of the anchor information handling system node **100**, via wired connections as applicable, or when AC power from the AC power adapter **126** is removed.

(35) In a particular non-limiting, exemplary embodiment, the computer-readable medium can include a solid-state memory such as a memory card or other package that houses one or more non-volatile read-only memories. Further, the computer-readable medium can be a random-access memory or other volatile re-writable memory. Additionally, the computer-readable medium can include a magneto-optical or optical medium, such as a disk or tapes or other storage device to store information received via carrier wave signals such as a signal communicated over a transmission medium. Furthermore, a computer readable medium **110** can store information received from distributed network resources such as from a cloud-based environment. A digital file attachment to an e-mail or other self-contained information archive or set of archives may be considered a distribution medium that is equivalent to a tangible storage medium. Accordingly, the disclosure is considered to include any one or more of a computer-readable medium or a distribution medium and other equivalents and successor media, in which data or machine-readable code instructions may be stored.

(36) In other embodiments, dedicated hardware implementations such as application specific integrated circuits (ASICs), programmable logic arrays and other hardware devices can be constructed to implement one or more of the methods described herein. Applications that may include the apparatus and systems of various embodiments can broadly include a variety of electronic and computer systems. One or more embodiments described herein may implement functions using two or more specific interconnected hardware modules or devices with related control and data signals that can be communicated between and through the modules, or as portions of an application-specific integrated circuit. Accordingly, the present system encompasses hardware resources executing software or firmware, as well as hardware implementations.

(37) As described herein, the anchor information handling system node **100** is operatively coupled to a peripheral device workspace cloud orchestrator **156** executing on a peripheral device workspace cloud orchestrator server **158** over a network that includes any number of servers, computing devices, and other cloud computing resources such as the peripheral device workspace cloud orchestrator server **158**. The peripheral device workspace cloud orchestrator **156** may, therefore, execute on and include any hardware of one or more peripheral device workspace cloud orchestrator servers **158** that may be distributed over multiple physical locations but act in concert with each other. The peripheral device workspace cloud orchestrator **156** is executed on the peripheral device workspace cloud orchestrator server **158** to facilitate, in an example embodiment, an ITDM to, via hardware device operational policies, cause a peripheral device workspace to be created that includes any number of device nodes (e.g., including the anchor information handling system node **100** as an anchor device node and a plurality of secondary nodes including smart device nodes and peripheral device nodes) forming part of the peripheral device workspace after receiving device enrollment data describing one or more device nodes. In an embodiment, the ITDM may also create hardware device operational policies that, based on the registered device nodes detected within the one or more created collaborative peripheral device workspaces, facilitate the operations of the various nodes within the collaborative peripheral device workspace. In an embodiment, the ITDM may cause hardware device operational policies to be applied to the created collaborative peripheral device workspace. It is appreciated that the device nodes described herein may include each of the peripheral device nodes operatively coupled to the anchor information handling system node **100** which acts as a primary or anchor information handling

system node **100** and the peripheral device workspaces created may be described and identified with a peripheral device workspace identification values. In an embodiment, the peripheral device workspace identification values may be stored on the cloud orchestrator database **188** by the peripheral device workspace cloud orchestrator server **158**.

(38) The peripheral device workspace cloud orchestrator server **158** may be any computing device that may include similar elements as the information handling system **100** such as a memory device, a cloud orchestrator hardware processing device **170**, a PMU, and other elements that allow the peripheral device workspace cloud orchestrator server **158** to execute computer-readable program code of the cloud manageability orchestrator module **168**, ecosystem manageability service module **170**, cloud orchestrator notification module **172**, telemetry ownership orchestrator module **174**, telemetry aggregator module **176**, workspace capability orchestrator module **178**, workspace context discovery module **180**, and other software or firmware as described herein.

(39) As described herein, the peripheral device workspace cloud orchestrator server **158** includes a computer-readable program code of a peripheral device workspace cloud manageability orchestrator module **168** that, when executed by the cloud orchestrator hardware processing device **170** of the peripheral device workspace cloud orchestrator server **158**, uses device enrollment data describing one or more device nodes to create a manifest of the device nodes within the peripheral device workspace. The execution of the peripheral device workspace cloud manageability orchestrator module **168** creates a manifest of device nodes at an identified physical location of the peripheral device workspace with the peripheral device workspace having a peripheral device workspace identification value stored at the cloud orchestrator database **188** and accessible to the peripheral device workspace cloud orchestrator server **158**. In the context of the present specification and in the appended claims, a peripheral device workspace may be an ecosystem of device nodes (e.g., including a location identifier and peripheral devices coupled to the anchor information handling system node **100**, a docking station **151**, etc.) connected to a primary or anchor device node such as the anchor information handling system node **100**. In an embodiment, a peripheral device workspace may also be defined with a peripheral device workspace identification value and part of a user composite peripheral device workspace identifier and a type-classification for the type of peripheral device workspace associated with each of the anchor information handling system node **100** and peripheral device nodes (e.g., input/output devices **142**) such that a user may have multiple peripheral device workspaces having peripheral device workspace identification values associated with the user based on the context and/or environment of each identified peripheral device workspace. For example, a user composite peripheral device workspace identifier may be used to define a first peripheral device workspace at a home office having a first peripheral device workspace identification value, a second peripheral device workspace at a work office having a second peripheral device workspace identification value, a third peripheral device workspace at a different location (e.g., a coffee shop) having a third peripheral device workspace identification value, and other peripheral device workspaces that can be defined by both the device nodes included within the peripheral device workspace and the location identifier of the physical location for the peripheral device workspace (e.g., defined by location data such as GPS data as well as network data) and having a peripheral device workspace identification value.

(40) The peripheral device workspace cloud orchestrator server **158** may also execute computer-readable program code of an ecosystem manageability service module **170**. Execution of the ecosystem manageability service module **168** causes the peripheral device workspace cloud orchestrator server **158** to apply the hardware device operational policies created by the ITDM at the peripheral device workspace cloud orchestrator console **160** to the created peripheral device workspace. In an embodiment, the peripheral device workspace cloud manageability orchestrator module **168** creates hardware device operational policies for each of the one or more node devices within the peripheral device workspace based on received hardware device operational policies

created by the ITDM. As described herein, the ecosystem manageability service module **168** may identify those peripheral devices that form part of the peripheral device workspace having a peripheral device workspace identifier value and may propagate those hardware device operational policies that apply to those peripheral devices. For example, where a plurality of peripheral device workspaces with a plurality of peripheral device workspace identifier values each include a specific wireless mouse that the ITDM has created a hardware device operational policy for, the execution of the computer-readable program code of the ecosystem manageability service module **168** causes the peripheral device workspace cloud orchestrator server **158** to send those hardware device operational policies to the appropriate peripheral device workspace (e.g., to each of the central, main, or anchor node devices) so that those hardware device operational policies may be passed to the wireless mice in each peripheral device workspace having a peripheral device workspace identifier value where such as specific mouse model or type is located.

(41) Still further, the cloud orchestrator hardware processing device **166** of the peripheral device workspace cloud orchestrator server **158** may execute computer-readable program code of a telemetry aggregator module **176**. Execution of the telemetry aggregator module **176** causes the peripheral device workspace cloud orchestrator server **158** to aggregate telemetry data and KPI data associated with the device nodes within the peripheral device workspace or plural peripheral device workspaces received by the peripheral device workspace cloud orchestrator server **158** from a telemetry aggregator sub-agent **184-1**, **184-2** executing on the anchor information handling system node **100**. This aggregation process includes, in an embodiment, adding metadata to the telemetry data that categorizes the telemetry data into categories such as the type of peripheral device the telemetry data has originated from, the type of telemetry data originating from that device node, current settings at that device node, potential settings at that device node, among others. It is appreciated that this environmental telemetry data may form a context in which the device nodes are operating in a peripheral device workspace. For example, a first device node may be a built-in video display device (e.g., at a laptop) and a second device node may be an external video display device. In this example embodiment, the telemetry data associated with the built-in video display device may include metadata that categorizes the resolution data of the environmental telemetry data from the built-in video display device, the current display settings of the telemetry data from the built-in video display device, and so on may be labeled and aggregated with display device environmental telemetry data. Similarly, the environmental telemetry data associated with the external display device may include metadata that categorizes the resolution data of the telemetry data from the external display device, the current display settings of the telemetry data from the external video display device, and so on may be labeled and aggregated with display device environmental telemetry data.

(42) The environmental telemetry data itself may include any data, in situ of each of the peripheral device nodes within the created peripheral device workspace. Some specific examples of environmental telemetry data may include, but are not limited to, Input/Output (I/O) operation load on the peripheral device nodes, memory usage the peripheral device nodes, storage usage the peripheral device nodes, CPU/hardware processor power utilization of the peripheral device nodes, connectivity, wired or wireless connectivity quality, application and/or hardware uptime and downtime of the peripheral device nodes, user information, application errors and exceptions, database accesses of each of the peripheral device nodes, latency, web log information, and database errors and warnings. This telemetry data may be generated at and received from each of the individual peripheral device nodes in a peripheral device workspace. In an embodiment, the device nodes such as the anchor information handling system node **100** may execute computer-readable program code of a telemetry aggregator sub-agent **184-1**, **184-2** to aggregate the environmental telemetry data from the device node and other device nodes for preparation and labelling of the environmental telemetry data to be sent to the peripheral device workspace cloud orchestrator server **158**. Again, this aggregation process not only includes the accumulation of the

environmental telemetry data from each peripheral device node, but also the categorization of that telemetry data by adding metadata that defines categories of the environmental telemetry data. In an embodiment, the anchor information handling system node **100** may, with the hardware processor **102**, execute computer-readable program code of an ecosystem manageability and telemetry sub-agent **186-1**, **186-2** to transmit this aggregated environmental telemetry data to the peripheral device workspace cloud orchestrator server **158**.

(43) In an embodiment, the cloud orchestrator hardware processing device **166** may execute the computer-readable program code of the telemetry aggregator module **176** to aggregate key performance indicator (KPI) data associated with the plurality of device nodes within the peripheral device workspace. These KPIs may include general performance KPIs, battery availability KPIs, application KPIs, audio KPIs, video KPIs, threshold battery power level KPIs, threshold temperature level KPIs, and processing resource level KPIs, memory resource level KPIs, network traffic level KPIs, or other KPIs. In an embodiment, these KPIs may be included as environmental telemetry data sent from the anchor information handling system node **100** about the anchor information handling system **100** and any smart device nodes having network access to the peripheral device workspace cloud orchestrator server **158**. As described herein, the execution of the telemetry aggregator module **176** may also associate metadata to the environmental telemetry data that categorizes the environmental telemetry data to include categorization of this KPI data.

(44) In an embodiment, the cloud orchestrator hardware processing device **166** executes computer-readable program code of a telemetry ownership orchestrator module **174** to map the environmental telemetry data received from the anchor information handling system node **100** to each device node in the peripheral device workspace and determine, based on the KPI data and device node capabilities, whether telemetry aggregation functions are to be transitioned, at least partially, to a smart device node within the peripheral device workspace that may have an independent network connection to the peripheral device workspace cloud orchestrator server **158**. As described herein, the creation of the peripheral device workspace via execution of the peripheral device workspace cloud manageability orchestrator module **168**, includes the receipt of capabilities associated with each device node within the peripheral device workspace. In an example embodiment, the peripheral device workspace may include the anchor information handling system node **100** that is operatively coupled directly to secondary peripheral device nodes such as a keyboard **146** and mouse **152** as well as to a docking station **151** acting as a smart device node within the same peripheral device workspace. The docking station **151** may also be operatively coupled further to secondary peripheral device nodes such as an external monitor **144**, a webcam (not shown), and an audio device such as a speaker or a wireless headset **153**. Typically, the anchor information handling system node **100** may handle all the environmental telemetry data collection and aggregation responsibilities even for the external monitor **144**, webcam, and wireless headset **153** that is directly operatively coupled to the docking station **151**. However, the hierarchy of this telemetry data collection may be changed from the anchor information handling system node **100** being solely responsible for this collection and aggregation to the anchor information handling system node **100** and docking station **151** both being responsible for the collection and aggregation of the telemetry data associated with their own respective secondary peripheral device nodes when necessary. The peripheral device workspace cloud orchestrator server **158**, in this example embodiment, may determine whether any offloading of environmental telemetry data collection and aggregation is necessary, based on the KPI values, as it relates to individual peripheral device nodes within the workspace such as hardware resource utilization, memory resource utilization, or network traffic levels of the anchor information handling system node reaching threshold levels that may affect quality of experience and operations. In such example embodiments, some or all of the collection of environmental telemetry data and aggregation functions for secondary peripheral device nodes may be offloaded to a smart device node that is less busy.

(45) The capabilities associated with and described in the aggregated environmental telemetry data

may include, for example, video input/video output capabilities of the device node, audio input/audio output capabilities of the device node, network communication capabilities of the device node, processing capabilities of the device node, data storage capabilities of the device node, and wired or wireless connection capabilities of the device node (e.g., available ports or traffic levels), existing executable software/firmware stored on the device node, sensors present at the device node, security protocols present at the device node, among other capabilities. For example, a peripheral device workspace may include an external webcam, an internal webcam on the anchor information handling system node **100**, a video display device **144**, and an external video display device **144**. Each of these device nodes may include hardware that defines their capabilities (via the environmental telemetry data aggregated therefrom) such as the external webcam having a microphone to capture audio input, a camera to capture video input, a universal serial bus cable currently inserted into a USB port on the anchor information handling system node **100**, and a wireless interface adapter that can wirelessly couple the external webcam to the anchor information handling system node **100** instead of using a wired connection. All of these capabilities may be identified by the peripheral device workspace cloud orchestrator server **158** via the environmental telemetry data and the aggregation thereof after receiving device identification data from the anchor information handling system node **100**. These capabilities may be mapped by the execution of a workspace context discovery module **180** to provide recommendations, based on the capabilities and a context in which the anchor information handling system node **100** is operating, indicating which functionalities should be switched from one device to another. For example, where the external webcam is removed from the anchor information handling system node **100** and, accordingly, from the peripheral device workspace, the execution of the workspace context discovery module **180** may provide a recommendation to switch those capabilities associated with the external webcam to the capabilities associated with the internal webcam. The mapped capabilities based on the aggregated environmental telemetry data may then be updated and provided to the anchor information handling system node **100** and smart node devices within the peripheral device workspace.

(46) In an embodiment, the cloud orchestrator hardware processing device **166** of the peripheral device workspace cloud orchestrator server **158** may execute computer-readable program code of a workspace capability orchestrator module **178** to, based on the KPI data and context associated with each device node within the peripheral device workspace, provide peripheral device workspace recommendations to optimize configurations of the device nodes within the peripheral device workspace. The systems and methods described herein, therefore, allow for the partial offloading of environmental telemetry data collection and aggregation from one device node (e.g., anchor information handling system node **100**) to another device node (e.g., smart device node) when KPI changes are detected as well that indicate one device node has reached threshold levels of hardware process utilization, memory utilization, or network traffic levels and switch the functionalities of collecting and aggregating environmental telemetry data for at least a portion of the node devices in a peripheral device workspace from one node device to another node device. This switch may occur when the context and KPIs indicate that, in order to maintain a quality of experience (QoE), for example, these functionalities should be switched.

(47) When referred to as a “system,” a “device,” a “module,” a “controller,” or the like, the embodiments described herein can be configured as hardware. For example, a portion of an information handling system device may be hardware such as, for example, an integrated circuit (such as an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a structured ASIC, or a device embedded on a larger chip), a card (such as a Peripheral Component Interface (PCI) card, a PCI-express card, a Personal Computer Memory Card International Association (PCMCIA) card, or other such expansion card), or a system (such as a motherboard, a system-on-a-chip (SoC), or a stand-alone device). The system, device, controller, or module can include hardware processing resources executing software, including firmware

embedded at a device, such as an Intel® brand processor, AMD® brand processors, Qualcomm® brand processors, or other processors and chipsets, or other such hardware device capable of operating a relevant software environment of the information handling system. The system, device, controller, or module can also include a combination of the foregoing examples of hardware or hardware executing software or firmware. Note that an information handling system can include an integrated circuit or a board-level product having portions thereof that can also be any combination of hardware and hardware executing software. Devices, modules, hardware resources, or hardware controllers that are in communication with one another need not be in continuous communication with each other, unless expressly specified otherwise. In addition, devices, modules, hardware resources, and hardware controllers that are in communication with one another can communicate directly or indirectly through one or more intermediaries.

(48) FIG. 2 is a graphic diagram of an anchor information handling system node **200** forming a peripheral device workspace and including a plurality of peripheral device nodes operatively coupled to the anchor information handling system according to another embodiment of the present disclosure. Further, the anchor information handling system node **200** is operatively coupled to a peripheral device workspace cloud orchestrator server **258** executing code instructions of one or more software modules for a peripheral device workspace cloud orchestrator **256** according to embodiments of the present disclosure. As described herein, the peripheral device workspace may include the anchor information handling system node **200** along with a plurality of other node devices. These node devices may include a mouse **252**, a wireless headset **253**, a stylus **248**, a trackpad **250**, a video display device **244-1**, an internal webcam **294-1**, a docking station **251**, a keyboard **246**, an external video display device **244-2**, a conference room video display device **244-3**, and an external webcam **294-2**. As shown in FIG. 2, the keyboard **246**, external video display device **244-2**, conference room video display device **244-3**, and external webcam **294-2** are operatively coupled to the docking station **251** with the docking station **251** coupled to the anchor information handling system **200**. This creates a hierarchy within the peripheral device workspace. Other secondary peripheral device nodes including mouse **252**, a wireless headset **253**, a stylus **248**, a trackpad **250**, a video display device **244-1**, an internal webcam **294-1** are operatively coupled directly with the anchor information handling system node **200**. It is appreciated that the peripheral device nodes shown in FIG. 2 are merely representative of a few potential peripheral device nodes that may be operatively coupled to the anchor information handling system **200** or through the docking station **251** and forming part of the peripheral device workspace. It is appreciated that other peripheral device workspaces may be formed that may include a different set of peripheral device nodes within those peripheral device workspaces.

(49) In the example shown in FIG. 2, the peripheral device workspace may include a smart device node such as a docking station **251**. As described herein, the docking station **251** may be identified as a smart device node due to the inclusion of a hardware processing device and/or network connection capabilities. This allows the docking station **251**, for example, to act as a telemetry aggregator instead of the anchor information handling system node **200** being solely operating as the environmental telemetry data collector and aggregator for the peripheral device workspace for data on nodes, their operation, their connectivity, and their power states among other environmental telemetry data. According to embodiments herein, the switch between the anchor information handling system node **200** being the only environmental telemetry data collector and aggregator to both the anchor information handling system node **200** and the docking station **251** acting as the environmental telemetry data collector and aggregator occurs when the KPI data and device node capabilities have indicated that the telemetry aggregation functions should be transitioned from the anchor information handling system node **200** to the docking station **251**. As described herein, this aggregation process may not only include the accumulation or collection of environmental telemetry data from each peripheral device node, but also include the categorization of that telemetry data by adding metadata that describes, for example, a type of peripheral device the

telemetry data has originated from, the type of telemetry data originating from that device node, current settings at that device node, potential settings at that device node, among others. It is appreciated that this data may form an environmental context in which the device nodes are operating in the peripheral device workspace. For example, a first device node may be a built-in video display device (e.g., at a laptop) and a second device node may be an external video display device. In this example, the environmental telemetry data associated with the built-in video display device may include metadata that categorizes the resolution data of the environmental telemetry data from the built-in video display device, the current display settings of the telemetry data from the built-in video display device, and so on for aggregation under display type environmental telemetry data. Similarly, the telemetry data associated with the external display device may include metadata that categorizes the resolution data of the environmental telemetry data from the external video display device, the current display settings of the environmental telemetry data from the external video display device, and so on for aggregation under display type environmental telemetry data. Additionally, as described herein, the peripheral device workspace cloud orchestrator server **258** may provide peripheral device workspace recommendations to optimize configurations of the device nodes within the peripheral device workspace based on the KPI data and context of operation of each device node within the peripheral device workspace. For example, environmental telemetry data categorized as display environmental telemetry data may be accessed to aggregate and compare display device settings for display device nodes in the peripheral device workspace.

(50) During operation, the execution of the ecosystem manageability service module **270** may be used to help create the peripheral device workspace. For example, the anchor information handling system node **300** may enroll a new device node (e.g., a wireless mouse) with a third-party manageability platform (e.g., FIG. 3, **397**) so that the cloud manageability orchestrator module **268** may access tokens sent from the third-party manageability platform. The ecosystem manageability and telemetry sub-agent **286-1** of the anchor information handling system node **200** may request that a workspace be created that includes the anchor information handling system node **200** and the new device node. As new device nodes are operatively coupled to the anchor information handling system node **200**, the execution of the cloud manageability orchestrator module **268** by the cloud orchestrator hardware processing device **266** of the peripheral device workspace cloud orchestrator server **258** may update a map and manifest of the node devices within the peripheral device workspace.

(51) With the peripheral device workspace being formed, the execution of a cloud manageability orchestrator sub-agent **296** by the hardware processing device **202** of the anchor information handling system **200** causes the anchor information handling system **200** to send device node capabilities associated with each device node in the peripheral device workspace including those peripheral devices associated with the anchor information handling system node **200** and the docking station **251**. Again, these capabilities may include for example, video input/video output capabilities of the device node, audio input/audio output capabilities of the device node, network communication capabilities of the device node, processing capabilities of the device node, data storage capabilities of the device node, and wired connection capabilities of the device node (e.g., available ports), existing executable software/firmware stored on the device node, sensors present at the device node, security protocols present at the device node, among other capabilities. In addition to the capabilities, the execution of the cloud manageability orchestrator sub-agent **296** causes the anchor information handling system **200** to send the KPIs and other telemetry data to the peripheral device workspace cloud orchestrator server **258** as described herein. Again, the KPI data may include processing performance KPI, battery availability KPI, audio KPIs, video KPIs, and the like. The telemetry data may include any data, in situ of each of the peripheral device nodes within the created peripheral device workspace. Some specific examples of telemetry data may include, but are not limited to, Input/Output (I/O) operation load on the peripheral device nodes, memory

usage the peripheral device nodes, storage usage the peripheral device nodes, CPU/hardware processor utilization, power utilization of the peripheral device nodes, wired or wireless connectivity quality, application and/or hardware uptime and downtime of the peripheral device nodes, user information, application errors and exceptions, database accesses of each of the peripheral device nodes, latency, web log information, and database errors and warnings. In an embodiment, the telemetry aggregator sub-agent **284-1** of the anchor information handling system node **200** may prepare this environmental telemetry data (including the KPI data) for aggregation prior to sending this data to the peripheral device workspace cloud orchestrator server **258** where it may be aggregated in various ways depending on execution of the one or more software modules of the peripheral device workspace cloud orchestrator **256**. Again, this aggregation process may include adding metadata to the environmental telemetry data that categorizes the environmental telemetry data into categories such as the type of peripheral device the environmental telemetry data has originated from, the type of environmental telemetry data originating from that device node, current settings at that device node, potential settings at that device node, among others. It is appreciated that this data may form an environmental context in which the device nodes are operating. For example, a first device node may be a built-in video display device **244-1** (e.g., at a laptop) and a second device node may be an external video display device **244-2** (e.g., operatively coupled to the docking station **251**). In this example, the environmental telemetry data associated with the built-in video display device **244-1** may include metadata that categorizes the resolution data of the environmental telemetry data from the built-in video display device **244-1**, the current display settings of the environmental telemetry data from the built-in video display device **244-1**, and so on as display device categorized environmental telemetry data. Similarly, the telemetry data associated with the external display device **244-2** may include metadata that categorizes the resolution data of the environmental telemetry data from the external video display device **244-2**, the current display settings of the environmental telemetry data from the external video display device **244-2**, and so on as display device categorized environmental telemetry data. This is just one example of types of categorized environmental telemetry data according to embodiments herein.

(52) This environmental telemetry data, after being prepared for aggregation, may be routed through the peripheral device server notification gateway **382** and to the peripheral device workspace cloud orchestrator server **258** executing computer-readable program code of the telemetry aggregator module **276**. The telemetry aggregator module **276** may aggregate the telemetry data. In an embodiment, this aggregated telemetry data may be used by the peripheral device workspace cloud orchestrator server **258**, executing a telemetry ownership orchestrator module **274**, to compute whether any telemetry offloading responsibilities are to take place such that at least some telemetry aggregation functions are offloaded from one device node (e.g., the anchor information handling system **200**) to another device node (e.g., the docking station **251**). The results of this determination may be retransmitted from the peripheral device workspace cloud orchestrator server **258** to the anchor information handling system **200** and any smart device node (e.g., docking station **251**) within the peripheral device workspace. For example, hardware processing utilization levels, memory utilization levels, or network congestion levels at the anchor node information handling system node **200** or the smart device node **251** may determine whether offloading of environmental telemetry gathering and aggregation processing should occur based on threshold levels of those factors. Based on the results of the determination by the telemetry ownership orchestrator module **274** whether offloading should occur, some of the environmental telemetry gathering and aggregation preparation processing may be offloaded or not to a second smart device node in the peripheral device workspace. In an embodiment, the anchor information handling system **200** may request that a mapping of the environmental telemetry gathering be made by the execution of the telemetry ownership orchestrator module **274** which will notify the anchor information handling system **200** and smart node devices which of the device nodes report to which

telemetry aggregator sub-agent **284-1**, **284-2** within the peripheral device workspace as well as which peripheral device nodes within the peripheral device workspace are responsible for collecting and aggregating the environmental telemetry data (e.g., the anchor information handling system node **200** or both the anchor information handling system node **200** and the docking station **251**).

(53) It is appreciated that the KPI data and environmental context data as part of the environmental telemetry data collected for a peripheral device workspace is used by the telemetry ownership orchestrator module **274** to determine whether some of the telemetry aggregation functions and responsibilities should be offloaded from one device node such as the anchor information handling system **200** to another device node such as the docking station **251**. Indeed, as the KPI data and environmental context data changes, the determination as to whether the aggregation functions should be offloaded may also change. For example, where the KPI data associated with the hardware processing device **202** of the anchor information handling system **200** indicates that a threshold hardware processing utilization limit has been reached or exceeded, this KPI data may be used in the calculations made by the telemetry ownership orchestrator module **274** to offload the telemetry aggregation data from the anchor information handling system **200** to, for example, the hardware processing device associated with the docking station **251**. However, where the KPI data has indicated that the hardware processing utilization threshold limit is no longer exceeded, the calculations made by the telemetry ownership orchestrator module **274** may indicate that the telemetry aggregation responsibilities are to be offloaded back to the hardware processing device **202** of the anchor information handling system **200**. This may be especially true when a hardware processing utilization threshold or other utilization threshold limit is reached at the docking station **251** in the interim.

(54) In another example, a change in environmental context data may also be used by the telemetry ownership orchestrator module **274** to determine whether to offload the telemetry aggregation responsibilities within the peripheral device workspace. For example, where the location of the peripheral device workspace has changed such that the docking station **251** is no longer operatively coupled to the anchor information handling system **200**, the environmental context data indicates this change such that offloading the telemetry aggregation responsibilities to the docking station **251** is no longer an option. However, other smart device nodes (if detected and available) within the new peripheral device workspace may be used to offload those telemetry aggregation responsibilities if determined from the environmental context data for the peripheral device workspace. Thus, the process of updating the telemetry aggregating device node within the peripheral device workspace may be a continual process based on the changing KPI data, context data, and capability data determined from the environmental telemetry data received from the peripheral device workspace.

(55) In an embodiment, the peripheral device workspace cloud orchestrator server **258** may execute computer-readable program code of a workspace context discovery module **280** to generate mapping recommendations that describe which capabilities **290-1** through **290-17** of which of the device nodes should be used within the peripheral device workspace. These capabilities may include, for example, video input/video output capabilities of the device node, audio input/audio output capabilities of the device node, network communication capabilities of the device node, processing capabilities of the device node, data storage capabilities of the device node, and wired connection capabilities of the device node (e.g., available ports), existing executable software/firmware stored on the device node, sensors present at the device node, security protocols present at the device node, among other capabilities. During operation, the anchor information handling system **200** may execute computer-readable program code of the workspace context discovery sub-agent **298** to discover node devices within the peripheral device workspace as well as the individual capabilities **290-1** through **290-17** associated with those node devices. Execution of the cloud manageability orchestrator sub-agent **298** causes the workspace context discovery sub-

agent **298** to send the specific capabilities **290-1** through **290-17** detected at each of the device nodes within the peripheral device workspace. The anchor information handling system **200** may also send the capabilities **290-1** through **290-17** to the workspace capability orchestrator module **278** of the peripheral device workspace cloud orchestrator server **258** for, in an example embodiment, storage on a cloud orchestrator database **288**.

(56) In an embodiment, execution of the cloud manageability orchestrator sub-agent **298** may cause the anchor information handling system **200** to request that the capabilities **290-1** through **290-17** be mapped. Additionally, the KPI data and environmental context data from gathered environmental telemetry data described herein is also sent with this request to the anchor information handling system **200**. This request is sent to the peripheral device workspace cloud orchestrator server **258** executing a workspace context discovery module **280** for this mapping of the capabilities **290-1** through **290-17**. As described herein, the execution of the cloud manageability orchestrator sub-agent **298** generates recommendations that map the capabilities **290-1** through **290-17** based on the context data and KPI data. For example, a peripheral device workspace may include an external webcam **294-2**, an internal webcam **294-1** on the anchor information handling system node **200**, a video display device **244-1**, and an external video display device **244-2**. Each of these device nodes may include hardware that defines their capabilities such as the external webcam **294-2** having a microphone to capture audio input, a camera to capture video input, a universal serial bus cable currently inserted into a USB port on the anchor information handling system node **200**, and a wireless interface adapter that can wirelessly couple the external webcam to the anchor information handling system node **200** instead of using a wired connection. All of these capabilities **290-1** through **290-17** may be identified by the peripheral device workspace cloud orchestrator server **258** after receiving device identification data from the anchor information handling system node **200**. These capabilities may be mapped by the execution of a workspace context discovery module **280** to provide recommendations, based on the capabilities **290-1** through **290-17** and a context indicated in environmental context data in which the anchor information handling system node **200** is operating, indicating which functionalities should be switched from one device to another. For example, where the external webcam **294-2** is removed from the anchor information handling system node **200** and, accordingly, from the peripheral device workspace, the execution of the workspace context discovery module **280** may provide a recommendation to switch those capabilities **290-1** through **290-17** associated with the external webcam **294-2** to the capabilities associated with the internal webcam **294-1**. The mapped capabilities may then be updated and provided to the anchor information handling system node **200** and smart node devices (e.g., docking station **251** within the peripheral device workspace).

(57) It is appreciated that, in one embodiment, the user of the anchor information handling system **200** may be notified of any offloading of the telemetry aggregation responsibilities within the peripheral device workspace and/or the switching between different capabilities **290-1** through **290-17** of one device node to another. In an embodiment, the peripheral device workspace cloud orchestrator server **258** may execute computer-readable program code of a cloud orchestrator notification module **272** to provide such notifications to the user during operation of the systems described herein.

(58) FIG. 3 is a flow chart showing a method **301** of executing code instructions of a cloud manageability orchestrator module **368** on a peripheral device workspace cloud orchestrator server **358** to automatically register node devices within one or more peripheral device workspaces to manage and orchestrate those node devices according to an embodiment of the present disclosure. FIG. 3 shows more details related to the execution of code instructions of a cloud manageability orchestrator module **368** for registering of those peripheral devices and anchor information handling system node **300** to the peripheral device workspaces having unique peripheral device workspace identification values and stored in a workspace database accessible by the cloud manageability orchestrator module **368** in order to allow an ITDM to manage those peripheral

device and information handling systems associated with one or more users.

(59) In an embodiment, at line **305**, the computer-readable program code of the ecosystem manageability sub-agent **386** is executed by a hardware processing device of the information handling system **300** in order to detect when a third-party manageability agent **395** executing on the information handling system **300** has enrolled a peripheral device or information handling system. This process of detecting when the third-party manageability agent **395** enrolls peripheral devices may be continuous in an embodiment. In another embodiment, the execution of the ecosystem manageability sub-agent **386** may periodically determine if and when the third-party manageability agent **395** has enrolled any peripheral devices as operational within a peripheral device workspace at line **310**.

(60) At line **315**, the cloud manageability orchestrator module **368** may log into one or more third-party manageability platforms **397** via the API for corresponding peripheral devices as described in embodiments herein. In an embodiment, the third-party manageability platform **397** and cloud manageability orchestrator module **368** may work in concert to provide to the cloud manageability orchestrator module **368** any enrollment data and peripheral device ID data associated with any enrolled peripheral devices in a peripheral device workspace.

(61) At line **320** an access token from the one or more third-party manageability platforms **397** may be provided to the cloud manageability orchestrator module **368**. This access token allows the cloud manageability orchestrator module **368** to gain access to the peripheral device data for each enrolled peripheral device and information handling system from the one or more third-party manageability platforms.

(62) At line **325**, the ecosystem manageability sub-agent **386** may create a peripheral device workspace by identifying those peripheral devices and information handling systems that are operated by a specific user, a location where those peripheral devices and information handling systems are being operated, and the association of any peripheral devices to the information handling system as collected by the ecosystem manageability sub-agent **386**. Such information may be used by the cloud manageability orchestrator module **368** executing at the peripheral device workspace cloud orchestrator server **358** for determination and identification of the peripheral device workspace with a peripheral device workspace identification value stored in a peripheral device workspace manifest at a workspace database associated with the peripheral device workspace cloud orchestrator server **358**. This allows the ecosystem manageability service module **370** to create a peripheral device workspace with the cloud manageability orchestrator module **368** at the peripheral device workspace cloud orchestrator server **358** based on this data at line **325**.

(63) The ecosystem manageability service module **386** may continue to identify this created peripheral device workspace at line **330** at the cloud manageability orchestrator module **368** with a unique peripheral device workspace identification value as described in embodiments herein. The execution of the computer-readable program code of the ecosystem manageability service module **386** may further associate the data for a manifest of nodes in the identified peripheral device workspaces from the ecosystem manageability sub-agent **386** by associating a service tag and peripheral device workspace identification value for each of the created peripheral device workspaces. In an embodiment, the association of a service tag and peripheral device workspace identification value to the created peripheral device workspace may be stored on a workspace database (not shown) associated with the peripheral device workspace cloud orchestrator server **358** as directed by the cloud manageability orchestrator module **368**.

(64) At line **335**, the ecosystem manageability sub-agent **386** may enumerate connected peripheral devices to the anchor information handling system node **300** acting as the primary or anchor node device and associated with the user. This enumeration of the connected peripheral device nodes may include the ecosystem manageability sub-agent **386** detecting which peripheral devices are operatively coupled to the information handling system **300**, how those peripheral device nodes are operatively coupled to the information handling system **300**, and an peripheral device identification

(ID) associated with each of the connected peripheral device nodes, among other identification data as well as a location for enumeration of peripheral devices and other nodes in a peripheral device workspace for association with a peripheral device workspace identification value. Additionally, the enumeration of the connected peripheral devices may also include assigning a user to each of the connected peripheral devices for a peripheral device workspace identification value with a composite peripheral device workspace identifier for a user such that those peripheral devices operatively coupled to the user's information handling system **300** are also tied to that user.

(65) In an embodiment, the execution of the computer-readable program code of the ecosystem manageability sub-agent **386** may then register, at line **340**, those enumerated peripheral device with the ecosystem manageability service module **370** in the peripheral device workspace associated with the particular peripheral device workspace identification value. This allows the ecosystem manageability service module **370** to determine the layout of each of the created peripheral device workspaces that forms a framework environmental context data for the peripheral device workspace and, at line **345**, create unique identifiers for each of the peripheral device nodes registered with a peripheral device workspace identification value from the ecosystem manageability sub-agent **386**. This unique identifier for each of the peripheral devices may be a random number, a serial number, or the like and may be included in a manifest for a peripheral device workspace identification value as well as associated with the composite peripheral device workspace identifier associated with the user. Thus, the ecosystem manageability service module **370** may be capable of specifically identifying a specific peripheral device within a specific created peripheral device workspace having a peripheral device workspace identification value and that is associated with a specific user, among potentially plural peripheral device workspaces for the user, via a composite peripheral device workspace identifier for a user.

(66) At line **350**, the computer-readable program code of the ecosystem manageability service module **370** may be executed by the hardware processor of the peripheral device workspace cloud orchestrator server **358** to update a peripheral device workspace having a peripheral device workspace identification value to include each peripheral device. As described in other example embodiments herein, certain peripheral devices may be swapped out for other peripheral devices causing the workspace to change context and environmental context data is changed. For example, a user may operatively couple a secondary or tertiary external monitor to a docking station coupled to the information handling system **600**. These new external monitors may be incorporated into the existing created peripheral device workspace with a particular peripheral device workspace identification value and the ecosystem manageability service module **370** may update that workspace as these new peripheral devices are discovered by the ecosystem manageability sub-agent **386**.

(67) This allows the ecosystem manageability service module **370** to, at line **355**, provide this data to the cloud manageability orchestrator module **368** so that the modification of the peripheral device workspace created by the information handling system **300** and its peripheral devices in a manifest associated with the particular peripheral device workspace identification value may be updated at a database (not shown) associated with the peripheral device workspace cloud orchestrator server **358**. These created, modified, and updated peripheral device workspaces having unique peripheral device workspace identification values may be saved on this database and may be associated with each user of each information handling system **300** in a composite peripheral device workspace identifier for a user having access within the plurality of created peripheral device workspaces.

(68) At line **361**, the cloud manageability orchestrator module **368**, an ITDM, or an automatic hardware device operational policy of an enterprise may search for the peripheral devices on a management database (not shown) associated with the third-party manageability platform **397** for peripheral devices associated with that third-party that are part of created or updated peripheral device workspace with a peripheral device workspace identification value. This management

database may include specific details about the enrolled peripheral devices at line **305** and **310** as well as general information about the capabilities of the type of peripheral devices enrolled and potential settings associated with each of the enrolled peripheral devices. The third-party manageability platform **397** may return this device-specific and general data to the cloud manageability orchestrator module **368** at line **365** if found. For example, a user may have been associated with a specific peripheral device workspace in a composite peripheral device workspace identifier for the user identifying that peripheral device workspace identification value, such as a home office peripheral device workspace, the peripheral device workspace identification value may be stored with a peripheral device workspace manifest that includes an information handling system **300** acting as the primary or anchor node device with a plurality of peripheral devices operatively coupled to the information handling system **300** associated with that user. It may arise that an ITDM may order a new peripheral device for the user such as a wireless mouse with the new wireless mouse being operatively coupled to the information handling system at the peripheral device workspace of the particular peripheral device workspace identification value, such as of the home office peripheral device workspace. In this example, the old mouse associated with the user's information handling system and indicated as being part of the created workspace may be removed from the particular peripheral device workspace, such as from the manifest of nodes of the peripheral device workspace identification value for the home office peripheral device workspace. As such, the cloud manageability orchestrator module **368** may modify the environmental context data for the created peripheral device workspace for the peripheral device workspace identification value of the home office peripheral device workspace to not include the old mouse and replace it with the new wireless mouse that has the device-specific and general data associated with it. The ecosystem manageability service module **370** executed on the peripheral device workspace cloud orchestrator server **358** may record this modification of the peripheral device workspace manifest associated with the peripheral device workspace identification value for the home office peripheral device workspace to indicate the new wireless mouse and its device-specific and general data as being part of the home office peripheral device workspace and the old mouse is no longer to be included in the home office peripheral device workspace.

(69) At line **375**, the cloud manageability orchestrator module **368** may update the workspace based on the device-specific and general data added and indicated as being associated with each of the new peripheral devices in the workspace. The device object is updated at the management database of the third-party manageability platform **397** for the peripheral device identification so that any updates or functional modifications may be retrieved from the third-party manageability platform **397** for that peripheral device if such should occur and managed and controlled by the execution of the cloud manageability orchestrator module **368** at the peripheral device workspace cloud orchestrator server **358**. At line **380**, the third-party manageability platform **397** may indicate whether the update of the workspace at the management database was successful or not to the cloud manageability orchestrator module **368** executing at the peripheral device workspace cloud orchestrator server **358**.

(70) It is appreciated as well that the ITDM may be able to access this created and modified peripheral device workspace by using the peripheral device workspace cloud orchestrator console **360** to access the peripheral device workspace from the cloud manageability orchestrator module **368** at line **385**. In an embodiment, the created and modified peripheral device workspace, such as the home office peripheral device workspace, is rendered with a manifest of peripheral device nodes identified, capabilities, connectivity options, and operational or update status as well as other peripheral device or node data on a graphical user interface (GUI) presented to the ITDM at a video display device of the peripheral device workspace cloud orchestrator console as described herein. Again, the ITDM may determine a number of potential issues associated with the modification of the workspace including, but not limited to, whether the new wireless mouse is compliant or not with current hardware device operational policies provided by the ITSM at the third-party

manageability platform **397**, whether proper firmware/software updates have been received at or propagated to the newly added wireless mouse, and whether the wireless mouse is also forming or should be part of a different workspace such as a hybrid work office peripheral device workspace. As such, the ITDM, via execution of the cloud manageability orchestrator module **368** on the peripheral device workspace cloud orchestrator server **358**, can access this data for any of a plurality of peripheral device workspaces in order to create and orchestrate one or more peripheral device workspaces.

(71) FIG. **4** is a flow chart showing a method **401** of computing telemetry offloading within a peripheral device workspace according to an embodiment of the present disclosure according to an embodiment of the present disclosure. As described herein, by offloading some portion of the telemetry aggregation responsibilities from a first device node **400** (e.g., anchor information handling system) to a second device node **403** (e.g., a smart device node such as a docking station) increases the quality of experience (QoE) of the peripheral device workspace for users by seamlessly and automatically reducing processing loads on a hardware processing devices within the anchor information handling system when changes in device capabilities, KPIs, and environmental context data changes as indicated by gathered environmental telemetry data from the peripheral device workspace.

(72) At line **402** execution of the cloud manageability orchestrator-sub-agent **496** may periodically send device node capabilities, KPIs, and environmental context data in environmental telemetry data collected for and associated with each of the device nodes in the peripheral device workspace. This environmental telemetry data may be collected and aggregated, at least initially, by a telemetry aggregator sub-agent **484-1** as described herein. This environmental telemetry data may include, for example, any data, in situ of each of the peripheral device nodes. Some specific examples of environmental context data in the environmental telemetry data may include, but are not limited to, Input/Output (I/O) operation load on the peripheral device nodes, memory usage the peripheral device nodes, storage usage the peripheral device nodes, CPU/hardware processor power utilization of the peripheral device nodes, connectivity of node devices, connectivity quality, application and/or hardware uptime and downtime of the peripheral device nodes, user information, application errors and exceptions, database accesses of each of the peripheral device nodes, latency, web log information, and database errors and warnings. The KPIs may be a category of environmental telemetry data that include general performance KPIs, battery availability KPIs, application KPIs, audio KPIs, video KPIs, threshold battery power level KPIs, threshold temperature level KPIs, and threshold processing resource level KPIs among other KPIs. Device capabilities data in the environmental telemetry data may describe video input/video output capabilities of the device node, audio input/audio output capabilities of the device node, network communication capabilities of the device node, processing capabilities of the device node, data storage capabilities of the device node, and wired connection capabilities of the device node (e.g., available ports), existing executable software/firmware stored on the device node, sensors present at the device node, security protocols present at the device node, among other capabilities. This capability data, KPI data, and environmental context data in collected environmental telemetry data may be processed for aggregation, at least initially, by the first device node **400** and transmitted to peripheral device workspace cloud orchestrator server **458** by way of a peripheral device server notification gateway **482** which, at line **404** relays the capability data, KPI data, and environmental context data in the environmental telemetry data to the peripheral device workspace cloud orchestrator server **458**. As described herein, this aggregation process may include adding metadata to the telemetry data that categorizes the environmental telemetry data into categories such as the type of peripheral device the environmental telemetry data has originated from, the type of environmental telemetry data originating from that device node, current settings at that device node, potential settings at that device node, among others. It is appreciated that this data may form a context in which the device nodes are operating in a peripheral device workspace. For example, a first device node may be a

built-in video display device (e.g., at a laptop) and a second device node may be an external video display device. In this example, the environmental telemetry data associated with the built-in video display device may include metadata that categorizes the resolution data of the telemetry data from the built-in video display device, the current display settings of the telemetry data from the built-in video display device, and so on that is in a display device category for collected environmental telemetry data. Similarly, the telemetry data associated with the external display device may include metadata that categorizes the resolution data of the environmental telemetry data from the external video display device, the current display settings of the environmental telemetry data from the external video display device, and so on that is also in a display device category for collected environmental telemetry data from the peripheral device workspace. This is just one example of a display device category for collected environmental telemetry data by which the environmental telemetry data may be aggregated ultimately at the peripheral device workspace cloud orchestrator server **458**.

(73) The peripheral device workspace cloud orchestrator server **458** executes computer-readable program code of a telemetry aggregator module **476** to, at line **406**, aggregate the capability data, KPI data, and environmental context data in the environmental telemetry data whether they come from two different device node communication links in the peripheral device workspace or aggregated based on one or more other categories of environmental telemetry data. In an embodiment, the execution of the telemetry aggregator module **476** may also store the capability data, KPI data, and environmental context data in the environmental telemetry data on a cloud orchestrator database and may associate the capability data, KPI data, and environmental context data in the environmental telemetry data with a user identification, peripheral device workspace identification, and device node identification as well as any other categorization applied.

(74) At line **408**, the aggregated capability data, KPI data, and environmental context data in the environmental telemetry data is provided and used by the peripheral device workspace cloud orchestrator server **458** executing computer-readable program code instructions of a telemetry ownership orchestrator module **474**. By executing the telemetry ownership orchestrator module **474**, the peripheral device workspace cloud orchestrator server **458** determines the telemetry aggregation offloading at line **410**. The peripheral device workspace cloud orchestrator server **458** determines whether, based on the capability data, KPI data, and environmental context data in the environmental telemetry data, some of the environmental telemetry collection and aggregation processing should be offloaded from the first device node **400** to the second device node **403** within the peripheral device workspace. The transitioning of some of these environmental telemetry collection and aggregation responsibilities from the first device node **400** to the second device node **403** may generally depend on the current status of hardware processing resources, memory resources, network connectivity usage levels or other usage levels of resources of the first device node **400**. For example, execution of code instructions of a telemetry ownership orchestrator module **474** may determine from the capability data the available processing resources available at the first device node **400**, a current load on the hardware processing device on the first device node **400** as well as current operation context within the peripheral device workspace such as whether processing-intensive applications are being executed by the hardware processing device within the first device node **400** reaching a hardware processing threshold level of usage. Further, execution of code instructions of a telemetry ownership orchestrator module **474** may determine that the capability data indicates the hardware processing capacity on other device nodes within the peripheral device workspace including the second device node **403** (e.g., smart device node) have greater availability and are not reaching a usage threshold level. For example, where the first device node **400** is initially responsible for aggregating the capability data, KPI data, and environmental context data in the environmental telemetry data for all the device nodes within the peripheral device workspace and the first device node **400** is executing a gaming application that requires a significant amount of hardware processing resources, this capability data, KPI data, and

environmental telemetry data is used by the telemetry ownership orchestrator module **474** to determine whether some of the environmental telemetry collection and aggregation processing responsibilities should be offloaded.

(75) At lines **412**, **414**, the success of the offloading determination from the execution of the code instructions of the telemetry ownership orchestrator module **474** may be sent to the cloud manageability orchestrator-sub-agent **496** via the peripheral device server notification gateway **482**. At lines **416** and **418**, the ecosystem manageability and telemetry sub-agent **486-1** of the first device node **400** and the ecosystem manageability and telemetry sub-agent **486-2** of the second device node **403** are notified regarding the results of the determination of the telemetry offloading at line **410**. This transmission instructs the first device node **400** and the second device node **403** concurrently about which of these device nodes is now responsible for telemetry collection and aggregation processing of environmental telemetry data from assigned or “owned” peripheral device nodes within the peripheral device workspace. It is appreciated that these environmental telemetry data collection and aggregation processing functions performed by any of the ecosystem manageability and telemetry sub-agents **486-1**, **486-2** may be preparation for those functions associated with the operation of the telemetry aggregator module **476** at the peripheral device workspace cloud orchestrator server **458** to ultimately aggregate the collected environmental telemetry data whether from one or multiple device nodes in a peripheral device workspace. Again, these aggregation processes include in some embodiments, adding metadata to the telemetry data that categorizes the telemetry data into categories such as the peripheral device workspace providing the environmental telemetry data, the type of peripheral device the environmental telemetry data has originated from, the type of environmental telemetry data originating from that device node, current settings at that device node, potential settings at that device node, among others. It is appreciated that this data may form a context in which the device nodes are operating within the peripheral device workspace.

(76) In an example, where the determination of offloading at line **410** by the execution of the code instructions of the telemetry ownership orchestrator module **474** has indicated that the telemetry aggregation responsibilities are to be transitioned from the first device node **400** to the second device node **403**, the hardware processing device of the first device node **400** may discontinue the execution of the telemetry aggregator sub-agent **484-1** and stop collecting and aggregating the capability data, KPI data, and environmental context data in the environmental telemetry data on behalf of a subset of device nodes within the peripheral device workspace that have been offloaded to the alternative smart device node. Concurrently, the second device node **403** is instructed to execute the ecosystem manageability and telemetry sub-agent **486-2** and a corresponding telemetry aggregator sub-agent to gather and process for aggregation the capability data, KPI data, and environmental context data in the environmental telemetry data from each device node that has been offloaded to the second device node **403** within the peripheral device workspace.

(77) At lines **420** and **422**, the ecosystem manageability and telemetry sub-agent **486-1** executing on the first device node **400** may periodically request for telemetry mapping via the peripheral device server notification gateway **482**. This request may be made to determine which device nodes within the peripheral device workspace the ecosystem manageability and telemetry sub-agent **486-1** is owned by which of the anchor information handling system node or the smart peripheral device node to collect and process for aggregation the capability data, KPI data, and environmental context data in the environmental telemetry data for aggregation at the peripheral device workspace cloud orchestrator server **458** depending on offloading determined at line **410**. Upon offloading of some portion of the node devices, the first device node **400** is no longer responsible for collecting or aggregating the capability data, KPI data, and environmental context data in the environmental telemetry data for those identified node devices owned, or operatively coupled as nodes secondary to the second device node **402** (e.g., a smart device node). The first device node **400** may be provided, by this request, with the identification of the device nodes the first device node **400** is

still required to collect and process for aggregation this capability data, KPI data, and telemetry data for. The response via the peripheral device server notification gateway **482** at lines **424** and **426** provides this device node identification or identifications for which the first device node **400** is to collect and process for aggregation the capability data, KPI data, and environmental context data in the environmental telemetry data. At line **428**, the ecosystem manageability and telemetry sub-agent **486-2** of the second device node **403** is also notified of the telemetry mapping for the portion of node devices it is now instructed to collect and process for aggregation the capability data, KPI data, and environmental context data in the environmental telemetry data.

(78) At line **430** and **432** the ecosystem manageability and telemetry sub-agent **486-1** of the first device node **400** may collect the capability data, KPI data, and environmental context data in the environmental telemetry data at its respective assigned device nodes and send it to the telemetry aggregator sub-agent **484-1**. Continuing with the example presented herein in FIG. 2, for example, the hardware processing device of the first device node **400** (e.g., anchor information handling system node) may execute the computer-readable program code of the ecosystem manageability and telemetry sub-agent **486-1** to collect the telemetry data associated with those peripheral device nodes that are assigned to it or owned by it. For example, those peripheral device nodes that are operatively coupled directly to the anchor information handling system node **400** (e.g., the mouse, the wireless headset, the stylus, the trackpad, and the video display device) may not be accessible by an alternative smart node device and, thus, is owned by the first device node **400** (e.g., anchor information handling system node).

(79) Execution of code instructions of the telemetry aggregator sub-agent **484-1** may operate to process the collected environmental telemetry data for aggregation at line **434** according to embodiments described herein for those node devices owned by the first device node **400** (e.g., anchor information handling system node). Because of the operations of the telemetry ownership orchestrator module **474**, some of the telemetry collection and aggregation responsibilities are offloaded to the second device node **403** while others remain with the first node device **400**. At line **434**, the telemetry data may be processed for aggregation by the execution of the computer-readable program code of the telemetry aggregator sub-agent **484-1** and sent, at lines **436** and **438**, to the telemetry aggregator module **474** at the peripheral device workspace cloud orchestrator server **458** via the peripheral device notification gateway **482**.

(80) As described herein, some of the telemetry data collection and aggregation is offloaded to the second device node **403** pursuant to the determination of offloading made at line **410**. As such, the second device node **403** (e.g., docking station **251**, FIG. 2) may execute code instructions of the ecosystem manageability and telemetry sub-agent **486-2** to collect the environmental telemetry data at line **440**, send it to the telemetry aggregator sub-agent **484-2** at line **442**. It is appreciated that this collection of environmental telemetry data is from those device nodes assigned to and owned by the second device node **403** by the offloading determination at line **410**. For example, those device nodes that are directly and operatively coupled to the second device node **403** (e.g., keyboard **246**, external video display device **244-2**, conference room video display device **244-3**, and external webcam **294-2**, FIG. 2) may thus be accessed and owned by the second device node **403** for collection of environmental telemetry data from those node devices. Thus, although the first device node **400** may no longer be responsible for aggregating all of the capability data, KPI data, and environmental context data in the environmental telemetry data, the second device node **403** is responsible for collecting that capability data, KPI data, and telemetry data. Then, at line **444**, the collected environmental telemetry data by the second node device **403** is processed for aggregation by execution of code instructions of the telemetry aggregator sub-agent **484-4**. This is done in accordance with embodiments herein to label the collected environmental telemetry for the peripheral device workspace and other categories. This aggregation processed environmental telemetry data is then also transmitted, at line **446**, to the telemetry aggregator module **476** via the peripheral device server notification gateway **482**. Execution of code instructions of the telemetry

aggregator module **476** at the peripheral device workspace cloud orchestrator server **458** then may aggregate with the collected environmental telemetry data from lines **436**, **438** and line **446** for the peripheral device workspace. This data may then be forwarded to the telemetry ownership orchestrator module **474** at line **448** as described herein to determine if any changes in the environmental telemetry data may apply to change the offloading status again.

(81) At line **450**, execution of the code instructions for the telemetry ownership orchestrator module **474** at the peripheral device workspace cloud orchestrator server **458** may again monitor for changes in resource utilization for hardware processing, memory resources, network connectivity levels or the like to again determine if any adjustment to the offloading of responsibilities for collection and aggregation processing of the environmental telemetry data among the device nodes in the peripheral device workspace is triggered as described in embodiments herein. For example, the execution of code instructions of the telemetry ownership orchestrator module **474** at the peripheral device workspace cloud orchestrator server **458** may determine from received KPI data that a change in resource utilization, such as hardware processing resource utilization, memory resource utilization, or network connectivity resource utilization at first device node **400** (e.g., the anchor information handling system node) has changed to fall below a second utilization threshold that may be same as the first utilization threshold used at line **410** or may be different lower resource utilization threshold level used to reset or re-adjust the offloading by the telemetry ownership orchestrator module **474** at the peripheral device workspace cloud orchestrator server **458** in various embodiments.

(82) It is appreciated that, regardless of which device node is executing the processes at lines **430/434** and **440/444** among the device nodes in the peripheral device workspace, the current capability data, KPI data, and environmental context data in the environmental telemetry data is periodically transmitted to the peripheral device workspace cloud orchestrator server **458**, via the peripheral device server notification gateway **482**, for the telemetry aggregator module **476** to aggregate and store this data as described herein. This method **401** allows for the periodic offloading of capability data, KPI data, and telemetry data aggregation responsibilities from one device node to another device node within the peripheral device workspace thereby staggering this responsibility when hardware processing resources at any given smart device node (e.g., anchor information handling system node and smart device node).

(83) FIG. 5 is a flow chart showing a method **501** of generating capability mapping recommendations within a peripheral device workspace according to an embodiment of the present disclosure. The systems and methods described herein allow for the management of capabilities within a peripheral device workspace where redundant capabilities exist. By mapping these capabilities and generating recommendations of which node device is to be responsible for certain capabilities within the peripheral device workspace the QoE is increased resulting in higher user satisfaction.

(84) The method **501** may include, at line **502**, the anchor information handling system node **500** executing computer-readable program code of the cloud manageability orchestrator sub-agent **596**. The execution of the cloud manageability orchestrator sub-agent **596** by the hardware processing device of the anchor information handling system node **500** may, in an embodiment, provide device enrollment data that describes the individual device nodes within the peripheral device workspace as detected by the anchor information handling system node **500**. This enrollment data may be associated with a device ID that may describe the make, model, type, and device serial number of each device node. In an embodiment, this data may be sent to the peripheral device workspace cloud orchestrator server **558** so that the peripheral device workspace may be created as described in connection with FIG. 3.

(85) At line **504**, the execution of the workspace context discovery sub-agent **598** by the hardware processing device of the anchor information handling system node **500** may cause the anchor information handling system node **500** to discover these node devices operatively coupled to the

anchor information handling system node **500** in a location and that form the peripheral device workspace. During this discovery process, the anchor information handling system node **500** may collect and identify capabilities data from gathered environmental telemetry data associated with each of the device nodes. Environmental telemetry data, including capabilities data, may be gathered by execution of code instructions for the ecosystem manageability and telemetry sub-agent **586-1** at the anchor information handling system node **500** according to embodiments herein. Again, capabilities data may include video input/video output capabilities of the device node, audio input/audio output capabilities of the device node, network communication capabilities of the device node, processing capabilities of the device node, data storage capabilities of the device node, and wired connection capabilities of the device node (e.g., available ports), existing executable software/firmware stored on the device node, sensors present at the device node, security protocols present at the device node, among other capabilities. These capabilities for the device nodes collected by the anchor information handling system node may be associated with the device ID for each peripheral device in an orchestrated device descriptor (ODD) by execution of code instructions of the workspace context discovery sub-agent. It is appreciated that any number of device nodes may have capabilities similar to other device nodes such that one capability of a device node may be selected over the same capability of another device node during execution of the code instructions of the workspace capability orchestrator module **578** at the peripheral device workspace cloud orchestrator server **558** according to the method **501** described herein.

(86) At line **506**, the cloud manageability orchestrator sub-agent **596** requests the capability data, such as the ODD for discovered node devices within the peripheral device workspace from the workspace context discovery sub-agent **598** and receives that data at line **508**. Once the cloud manageability orchestrator sub-agent **596** has received the capability data defining the capabilities of each device node within the peripheral device workspace, the cloud manageability orchestrator sub-agent **596** sends the device capability data to the peripheral device workspace cloud orchestrator server **558** via a peripheral device server notification gateway **582** at lines **510** and **512**. This may include the location and manifest of device nodes in the peripheral device workspace as well as the ODD including capabilities for each node device in the peripheral device workspace. A notification of the delivery success of the capability data and ODD to the peripheral device workspace cloud orchestrator server **558** is also provided, via the peripheral device server notification gateway **582**, to the anchor information handling system node **500** at lines **514** and **516**.

(87) At line **518**, the execution of the computer-readable program code of the cloud manageability orchestrator sub-agent **596** causes the anchor information handling system node **500** to request that capabilities associated with each device node be mapped. Along with this request, the anchor information handling system node **500** may send environmental context data and KPI data from environmental telemetry data collected by the ecosystem manageability and telemetry sub-agent **568-1** to the peripheral device workspace cloud orchestrator server **558** via the peripheral device server notification gateway **582**. As described herein, the KPI data may be a category of environmental telemetry data that includes general performance KPIs, battery availability KPIs, application KPIs, audio KPIs, video KPIs, threshold battery power level KPIs, threshold temperature level KPIs, and threshold processing resource level KPIs among other KPIs. Additionally, environmental context data may include any environmental telemetry data that describes the operating environment of the peripheral device workspace such as physical location, device node enrollment data, current computer-readable program code being executed by the hardware processing device of the anchor information handling system node **500** (e.g., gaming applications, videoconferencing applications, etc.), activities engaged in by the anchor information handling system node **500**, among other operational contexts.

(88) At line **522**, the hardware processing device of the peripheral device workspace cloud orchestrator server **558** executes computer-readable program code instructions of a workspace context discover module **580**. The execution of the code instructions the workspace context

discover module **580** causes the peripheral device workspace cloud orchestrator server **558** to generate mapping recommendations that describe which capabilities of which of the device nodes should be used by the anchor information handling system node within the peripheral device workspace. Again, the recommendations for mapped capabilities may depend on the operational context of the anchor information handling system node **500** as well as the KPI data received from the anchor information handling system node **500**. For example, where the context data indicates that the user is currently executing a gaming application and is using a wireless headset to communicate with other users online, the context data may indicate that the best option for audio input is not a microphone of an internal webcam on the anchor information handling system node **500** but instead is the microphone present on the wireless headset. Control over capabilities, such as overlapping capabilities, among plural peripheral device nodes is recommended for execution by the anchor information handling system node **500** at the peripheral device workspace. As such, the mapping recommendation may indicate that the microphone for the internal webcam is to be muted or otherwise disabled while the microphone for the wireless headset is mapped as the primary audio input sensor to be used within the peripheral device workspace. The opposite may be mapped by the peripheral device workspace cloud orchestrator server **558** where the context indicates that a videoconferencing application is being executed on the anchor information handling system node **500** and the internal webcam is being used to provide video input during the videoconferencing session. In this example, the microphone for the wireless headset is to be muted while the microphone associated with the internal webcam is to be mapped as being the primary audio input device within the peripheral device workspace. This context may again change where, during the videoconferencing session, the wireless headset detects that the user has placed the wireless headset on the user's head. At this point, because the context has changed by the user wearing the wireless headset, the recommendation for mapping the capabilities of the device nodes may be provided such that the microphone associated with the internal webcam is not muted or otherwise disabled and the microphone of the wireless headset is now to be used as the primary audio input device within the peripheral device workspace. It is appreciated that detected KPIs may also be used during this mapping process such that processing performance KPIs, battery availability KPIs, audio KPIs, video KPIs, and the like are used to determine whether certain device nodes are capable of performing the functions required by the user in specific operational environmental context data in various operating contexts of the anchor information handling system node **500**.

(89) Having generated the mapping recommendations at line **522**, the peripheral device workspace cloud orchestrator server **558** may send the mapped capabilities, based on the current environmental context data and KPI data, to the anchor information handling system node **500** via the peripheral device server notification gateway **582** at lines **524** and **526**. In an embodiment, the execution of the cloud manageability orchestrator sub-agent **596** may cause these mapped capabilities to be cached or otherwise stored on the anchor information handling system node **500** in a data storage device of the anchor information handling system node **500** at line **528** with a successful caching response being received at line **530**.

(90) At line **532**, the method **501** includes the anchor information handling system node **500** executing computer-readable program code of the ecosystem manageability and telemetry sub-agent **586-1** to assign the capabilities to be used by the anchor information handling system node **500** from particular device nodes in the peripheral device workspace based on the recommendations received from the peripheral device workspace cloud orchestrator server **558**. Again, this may cause the anchor information handling system node **500** to assign audio input responsibility to a specific microphone of a specific device node while disabling all other microphones associated with all other device nodes within the peripheral device workspace. This process may be carried out for each detected capability and for each device node within the peripheral device workspace.

(91) At line **534**, execution of the computer-readable program code of the ecosystem manageability

and telemetry sub-agent **586-1** causes the anchor information handling system node **500** to collect the environmental telemetry data from the discovered device nodes in the peripheral device workspace and monitor for changes in environmental context data and KPIs within the peripheral device workspace. For example, a change in environmental context data may include the user exiting a gaming application or executing a videoconferencing application at the hardware processing device of the anchor information handling system node **500**. Example changes in KPIs may include battery depletion of a rechargeable battery within a wireless peripheral device currently being used by the user to engage with the anchor information handling system node **500**. The present specification that multiple other types of changes to the operational environmental context data and KPIs associated with the operation of the anchor information handling system node **500** may be monitored by the anchor information handling system node **500** via execution of the ecosystem manageability and telemetry sub-agent **586-1**.

(92) At line **536**, any changes to the operational environmental context data and/or KPIs of the anchor information handling system node **500** may trigger remapping of the capabilities within the peripheral device workspace. During this time and in order to maintain a QoE, operational contexts and KPIs that change the QoE may cause the cloud manageability orchestrator sub-agent **596** to reassign capabilities by, for example, reenabling certain capabilities at certain device nodes while disabling certain capabilities at other device nodes. This reassignment may be temporary such that after the cloud manageability orchestrator sub-agent **596** receives a notification that the reassignment was successful at line **534**, the cloud manageability orchestrator sub-agent **596** again requests that that capabilities associated with each device node be mapped based on the detected changes to the operational context data and KPIs. This request at line **536** and **538** is relayed through the peripheral device server notification gateway **582** to the workspace context discover module **580** for the workspace context discover module **580** to update the mapping and return a notification at lines **540** and **542** that the remapping was successful.

(93) FIG. **6** is a flow chart showing method **600** of staggering telemetry for node capabilities management across device nodes in a peripheral device workspace according to an embodiment of the present disclosure. The method **600** described in FIG. **6** may allow for the capability data, KPI data, and environmental context data from environmental telemetry data gathered and prepared for aggregation by the most appropriate device node within the peripheral device workspace and proper assignment of device capabilities to the appropriate device node based on current and detected changes to the operational context data and KPI data for the anchor information handling system node or other smart device node gathering the environmental telemetry data from other device nodes in the peripheral device workspace.

(94) At block **605**, the method **600** may include sending the capability data, KPI data, and environmental context data in the environmental telemetry data gathered from device nodes at the peripheral device workspace to the peripheral device workspace cloud orchestrator server. This environmental telemetry data may include, for example, any data, in situ of each of the peripheral device nodes. Some specific examples of environmental context data in the environmental telemetry data may include, but are not limited to, Input/Output (I/O) operation load on the peripheral device nodes, memory usage the peripheral device nodes, storage usage the peripheral device nodes, CPU/hardware processor power utilization of the peripheral device nodes, connectivity of the device nodes, connectivity quality, application and/or hardware uptime and downtime of the peripheral device nodes, user information, application errors and exceptions, database accesses of each of the peripheral device nodes, latency, web log information, and database errors and warnings. The KPIs in the environmental telemetry data may be a category of environmental telemetry data that include general performance KPIs, battery availability KPIs, application KPIs, audio KPIs, video KPIs, threshold battery power level KPIs, threshold temperature level KPIs, and threshold processing resource level KPIs among other KPIs. Device capabilities data in the environmental telemetry data describe video input/video output capabilities

of the device node, audio input/audio output capabilities of the device node, network communication capabilities of the device node, processing capabilities of the device node, data storage capabilities of the device node, and wired connection capabilities of the device node (e.g., available ports), existing executable software/firmware stored on the device node, sensors present at the device node, security protocols present at the device node, among other capabilities. This capability data, KPI data, and environmental context data in the environmental telemetry data may be prepared for aggregation, at least initially, by the anchor information handling system node and transmitted to peripheral device workspace cloud orchestrator server by way of a peripheral device server notification gateway which relays the capability data, KPI data, and environmental context data in the environmental telemetry data to the peripheral device workspace cloud orchestrator server.

(95) At block **610**, the peripheral device workspace cloud orchestrator server executes computer-readable program code of a telemetry aggregator module to aggregate the capability data, KPI data, and environmental context data in the environmental telemetry data. In an embodiment, the execution of the telemetry aggregator module may also store the capability data, KPI data, and environmental context data in the environmental telemetry data on a cloud orchestrator database and may associate the capability data, KPI data, and environmental context data in the environmental telemetry data with a user identification, peripheral device workspace identification, and device node identification.

(96) At block **615**, the peripheral device workspace cloud orchestrator server executes computer-readable program code of a telemetry ownership orchestrator module to determine the telemetry aggregation offloading when a utilization level threshold of the anchor information handling system node or another smart device node is exceeded. This determination allows the peripheral device workspace cloud orchestrator server to determine whether, based on the capability data, KPI data, and environmental context data in the environmental telemetry data, the telemetry aggregation responsibilities should be offloaded from the anchor information handling system node to a smart device node (e.g., a docking station) within the peripheral device workspace. The transitioning of this responsibility from the anchor information handling system node to the smart device node may generally depend on the current status of hardware processing resources, memory resources, network connectivity conditions or other utilization levels of the anchor information handling system node. For example, the environmental telemetry data in capability data may indicate the available hardware processing resources available at the anchor information handling system node based on current load on the hardware processing device on the anchor information handling system node as well as current operational context within the peripheral device workspace such as whether processing-intensive applications are being executed by the hardware processing device within the anchor information handling system node. Further, the capability data may indicate hardware processing capacity on other device nodes within the peripheral device workspace including the smart device node. For example, where the anchor information handling system node is initially responsible for aggregating the capability data, KPI data, and environmental context data in the environmental telemetry data for all the device nodes within the peripheral device workspace and the anchor information handling system node is executing a gaming application that requires a significant amount of hardware processing resources, this capability data, KPI data, and environmental context data in the environmental telemetry data gathered for the peripheral device workspace is used by the telemetry ownership orchestrator module to determine whether the telemetry aggregation responsibilities should be offloaded due to the hardware processing device of the anchor information handling system node being oversubscribed when completing this task. Similarly, memory utilization levels and network connectivity connections such as congestion, bandwidth or throughput limitations, wired conditions, or radio conditions may also be used during execution of the telemetry ownership orchestrator module for determination of offloading of environmental telemetry data gathering and processing for aggregation in the peripheral device

workspace.

(97) At block **620**, the execution of the telemetry ownership orchestrator module determines whether the telemetry aggregation responsibilities should be offloaded from the anchor information handling system node to another smart device node within the peripheral device workspace. Where offloading of the telemetry data to another smart device node within the peripheral device workspace at block **620** is recommended, the method **600** continues to block **625** execution of code instructions of the telemetry ownership orchestrator module instructs the anchor information handling system node and smart device node to offload some portion of the environmental telemetry data collection and aggregation preparation responsibilities to the smart device node in an embodiment. Further, execution of the telemetry ownership orchestrator module may provide a response to the anchor information handling system node and the smart device node that the environmental telemetry data collection and aggregation preparation responsibilities are being offloaded to the smart device node in an embodiment. Where environmental telemetry data gathering and aggregation preparation responsibilities are not to be offloaded at block **620**, the method **600** may continue to block **630**.

(98) At block **630**, the peripheral device workspace cloud orchestrator server executes computer-readable program code of the workspace context discovery module to discover which node devices are operatively coupled to the anchor information handling system node that form the peripheral device workspace. During this discovery process, the anchor information handling system node may identify capabilities data associated with each of the device nodes within collected environmental telemetry data. Again, capabilities data may include video input/video output capabilities of the device node, audio input/audio output capabilities of the device node, network communication capabilities of the device node, processing capabilities of the device node, data storage capabilities of the device node, and wired connection capabilities of the device node (e.g., available ports), existing executable software/firmware stored on the device node, sensors present at the device node, security protocols present at the device node, among other capabilities. It is appreciated that any number of device nodes may have capabilities similar to other device nodes such that one capability of a device node may be selected over the same capability of another device node during the method **501** described herein. The capabilities data of device nodes in the peripheral device workspace may be utilized in execution of a workspace compatibility module of the peripheral device workspace cloud orchestrator to manage selection of peripheral device nodes to be used within a peripheral device workspace or for when a new peripheral device node is introduced or an old peripheral device is removed from the peripheral device workspace, in some example embodiments.

(99) In an embodiment, the discovery of which node devices are operatively coupled to the anchor information handling system node that form the peripheral device workspace with identified capabilities data associated with each of the device nodes within collected environmental telemetry data may be used to determine a smart device node for offloading the collection and aggregation processing of environmental telemetry data from one or more peripheral device nodes within the peripheral device workspace. This capabilities data and environmental context data may indicate to the executing telemetry ownership orchestrator module at the peripheral device workspace cloud orchestrator server, which device node or device nodes have hardware processing resources, memory resources, or networking capabilities for offloading collection and aggregation processing of gathered environmental telemetry data from at least a portion of device nodes within the peripheral device workspace.

(100) Execution of code instructions for the workspace context discovery module at block **630** also causes the peripheral device workspace cloud orchestrator server to generate mapping recommendations that describe which capabilities of which of the device nodes should be used within the peripheral device workspace including describing which peripheral device nodes are owned by a smart device node capable of receiving offloading and which are owned only by the

anchor information handling system node. Again, the recommendations for mapped capabilities may depend on the operational, environmental context data of the anchor information handling system node as well as the KPI data received from the anchor information handling system node in the first instance. For example, where the environmental context data indicates that the user is currently executing a gaming application and is using a wireless headset to communicate with other users online, the environmental context data may indicate that the best option for audio input is not a microphone of an internal webcam on the anchor information handling system node but instead is the microphone present on the wireless headset. In such an example embodiment, the mapping recommendation of execution of the workspace context discovery module may indicate that the microphone for the internal webcam is to be muted or otherwise disabled while the microphone for the wireless headset is mapped as the primary audio input sensor to be used within the peripheral device workspace. The opposite may be mapped by the peripheral device workspace cloud orchestrator server where the environmental context data indicates that a videoconferencing application is being executed on the anchor information handling system node and the internal webcam is being used to provide video input during the videoconferencing session. In this example, the microphone for the wireless headset is to be muted while the microphone associated with the internal webcam is to be mapped as being the primary audio input device within the peripheral device workspace. This environmental context data may again change where, during the videoconferencing session, the wireless headset detects that the user has placed the wireless headset on the user's head. At this point, because the environmental context data has changed by the user wearing the wireless headset, the recommendation for mapping the capabilities of the device nodes may be provided such that the microphone associated with the internal webcam is not muted or otherwise disabled and the microphone of the wireless headset is now to be used as the primary audio input device within the peripheral device workspace. It is appreciated that detected KPIs may also be used during this mapping process such that processing performance KPIs, battery availability KPIs, audio KPIs, video KPIs, and the like are used to determine whether certain device nodes are capable of performing the functions required by the user in specific operational contexts of the anchor information handling system node.

(101) Further, the recommendations for mapped capabilities may depend on the operational, environmental context data of the anchor information handling system node as well as the KPI data received from the anchor information handling system node as well as any smart device nodes that may be available for offloading of environmental telemetry data collection and aggregation processing. For example, where the environmental context data indicates that the user is currently executing a gaming application or videoconferencing application and is using a peripheral device node such as a wireless headset to communicate with other users online, the environmental context data may indicate that the anchor information handling system node is operatively coupled directly to this peripheral device node, but other peripheral device nodes are operatively coupled through a smart device node such as a smart dock to the anchor information handling system node. In such a case, the execution of code instructions for the workspace context discovery module may instead provide for mapped capabilities of the environmental context data for the smart node device and a portion of operatively coupled peripheral device nodes that it may “own” or administer to collection and aggregation processing of environmental telemetry data for a portion of the peripheral device workspace. Upon execution of code instructions of the telemetry ownership orchestrator module to determine that the processing resources, memory resources, or network connectivity has reached threshold utilization levels, instructions may be generated to the anchor information handling system and the identified smart device node to offload environmental telemetry data gathering and aggregation processing for some of the node devices that may be owned by the smart device node to ease the utilization burden on the anchor information handling system node. This will preserve the QoE for the user of the anchor information handling system node in the peripheral device workspace.

(102) At block **635**, the mapped capabilities, based on the current context and KPI data, are sent to the anchor information handling system node. In an embodiment, the execution of the cloud manageability orchestrator sub-agent may cause these mapped capabilities to be cached or otherwise stored on the anchor information handling system node in a data storage device of the anchor information handling system node such that an offload instruction from execution of code instructions of the telemetry ownership orchestrator module will indicate which smart device node will accommodate collection and aggregation processing of environmental telemetry data from a particular portion of node devices in the peripheral device workspace.

(103) At block **640**, the gathered environmental telemetry data with aggregation processing is received at the peripheral device workspace orchestrator server from one or both of the anchor information handling system node or smart device node depending on offloading status from the peripheral device workspace. Upon delivery of this aggregation processed environmental telemetry data from a combination of the anchor information handling system node for some portion of owned device nodes and separately via another network link from the smart node device for other node devices it owns, received environmental telemetry data may be aggregated at the peripheral device workspace orchestrator server for the peripheral device workspace. The execution of code instructions of the telemetry aggregator module at the peripheral device workspace orchestrator server may aggregate the received environmental telemetry data as sourced from the same peripheral device workspace and may further aggregate the environmental telemetry data based on other categorizations indicated in metadata added by the anchor information handling system node or the smart device node according to embodiments herein. Further at **640**, the anchor information handling system node may monitor for any changes in environmental context data, KPI data, or capability data in the collected environmental telemetry data to determine whether a further adjustment to the offloading of the telemetry data aggregation should occur and/or whether the capabilities should be remapped. Where changes in the capability data, KPI data, and environmental context data in the environmental telemetry data is detected by execution of code instructions of the telemetry ownership module at the peripheral device workspace orchestrator server, the method **600** may return to block **615** as described herein to determine from the received environmental telemetry data with the indicated changes whether threshold utilization levels have changed at the anchor information handling system node or one or more smart device nodes to again adjust offloading of the environmental telemetry data collection and aggregation processing from one or more node devices back to the anchor information handling system node or even to an additional smart device node with networking capabilities. Where, however, no changes in the capability data, KPI data, and environmental context data in the environmental telemetry data are detected, the method **600** may end although continuous monitoring for changes may occur and the process may start over.

(104) The blocks of the flow diagrams of FIGS. **3-6** or steps and aspects of the operation of the embodiments herein and discussed herein need not be performed in any given or specified order. It is contemplated that additional blocks, steps, or functions may be added, some blocks, steps or functions may not be performed, blocks, steps, or functions may occur contemporaneously, and blocks, steps, or functions from one flow diagram may be performed within another flow diagram.

(105) Devices, modules, resources, or programs that are in communication with one another need not be in continuous communication with each other, unless expressly specified otherwise. In addition, devices, modules, resources, or programs that are in communication with one another can communicate directly or indirectly through one or more intermediaries.

(106) Although only a few exemplary embodiments have been described in detail herein, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the embodiments of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the embodiments of the present disclosure as defined in the following

claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures.

(107) The subject matter described herein is to be considered illustrative, and not restrictive, and the appended claims are intended to cover any and all such modifications, enhancements, and other embodiments that fall within the scope of the present invention. Thus, to the maximum extent allowed by law, the scope of the present invention is to be determined by the broadest permissible interpretation of the following claims and their equivalents and shall not be restricted or limited by the foregoing detailed description.

Claims

1. A method of executing computer-readable program code of a telemetry ownership orchestrator module at a peripheral device workspace cloud orchestrator server for managing environmental telemetry data collection within a peripheral device workspace managed by the peripheral device workspace cloud orchestrator server comprising: executing computer-readable program code of a telemetry aggregator module at a peripheral device workspace cloud orchestrator server with a hardware processing device to receive and aggregate environmental telemetry data having KPI data associated with device nodes within the peripheral device workspace, the KPI data in the environmental telemetry data received, via a network interface device, from an anchor information handling system node among the device nodes within the peripheral device workspace along with device node capabilities associated with each device node; executing computer-readable program code of the telemetry ownership orchestrator module to map the environmental telemetry data received from the anchor information handling system node to each device node in the peripheral device workspace and determine, based on the KPI data and device node capabilities, whether telemetry data collection and aggregation functions are to be transitioned from the anchor information handling system node to a smart device node within the peripheral device workspace; and executing the computer-readable program code of the telemetry ownership orchestrator module to determine that a resource utilization level at the anchor information handling system has reached a first utilization threshold or that a processing-intensive application is being executed on the anchor information handling system node and transmitting a command to the peripheral device workspace to transition the telemetry data collection and aggregation functions from the anchor information handling system node to the smart device node.

2. The method of claim 1 further comprising: the hardware processor to execute computer-readable program code of a cloud manageability orchestrator module to register each of the plurality of device nodes within the peripheral device workspace and create a manifest of the device nodes for storage with a peripheral device workspace identification value at a database operatively coupled to the peripheral device workspace cloud orchestrator server.

3. The method of claim 1 further comprising: the hardware processor to execute computer-readable program code of an ecosystem manageability service module to apply the hardware device operational policies to the peripheral device workspace and identify those peripheral devices that form part of the peripheral device workspace having a peripheral device workspace identifier value and may propagate those hardware device operational policies that apply to those peripheral devices.

4. The method of claim 1, further comprising: receiving, via the network interface device, the environmental telemetry data including device node capability data and key performance indicator (KPI) data that includes hardware processing resource utilization at the anchor information handling system node; and executing the computer-readable program code of the telemetry ownership orchestrator module to determine that the hardware processing resource utilization is the resource utilization level that has reached the first utilization threshold for a processing resource of

the anchor information handling system node.

5. The method of claim 1 further comprising: receiving, via the network interface device, the environmental telemetry data including device node capability data and key performance indicator (KPI) data that includes memory resource utilization at the anchor information handling system node; and executing the computer-readable program code of the telemetry ownership orchestrator module to determine that the memory resource utilization is the resource utilization level that has reached the first utilization threshold for a memory of the anchor information handling system node.

6. The method of claim 1 further comprising: receiving, via the network interface device, the environmental telemetry data including device node capability data and key performance indicator (KPI) data that includes network connectivity resource utilization at the anchor information handling system node; and executing the computer-readable program code of the telemetry ownership orchestrator module to determine that the network connectivity resource utilization is the resource utilization level that has reached the first utilization threshold for a network interface device of the anchor information handling system node.

7. The method of claim 1 further comprising: receiving, via the network interface device, the environmental telemetry data including device node capability data and environmental telemetry data that includes data indicating that the processing-intensive application is being executed on the anchor information handling system node.

8. The method of claim 1 further comprising: executing computer-readable program code of a workspace context discovery module to monitor for changes in the environmental telemetry data within a peripheral device workspace; and transitioning, based on the changes in environmental telemetry data, the portion of environmental telemetry data collection and aggregation functions from the smart device node within the peripheral device workspace back to the anchor information handling system node when the resource utilization level at the anchor information handling system node is determined to have fallen below a second utilization threshold.

9. A peripheral device workspace cloud orchestrator server, comprising: a hardware processor; a memory device; a power management unit (PMU) to provide power to the hardware processor and memory device; a network interface device to receive environmental telemetry data including device node capability data and key performance indicator (KPI) data associated with a plurality of device nodes within a peripheral device workspace that includes an anchor information handling system node operatively coupled to a plurality of peripheral device nodes at a location; the hardware processor to execute computer-readable program code of a telemetry aggregator module to aggregate the environmental telemetry data collected and received from the device nodes within the peripheral device workspace; the hardware processor to execute computer-readable program code of a telemetry ownership orchestrator module to map the environmental telemetry data received from the anchor information handling system node and each peripheral device node of the plurality of peripheral device nodes in the peripheral device workspace and determine, based on the KPI data and device node capabilities, to offload a portion of environmental telemetry data collection and aggregation functions to be transitioned to a smart device node within the peripheral device workspace from the anchor information handling system node when a resource utilization level at the anchor information handling system node is determined to reach a utilization threshold; and the hardware processor to execute the computer-readable program code of the telemetry ownership orchestrator module to transmit instructions to the anchor information handling system node to offload the portion of environmental telemetry data collection and aggregation functions for a subset of the plurality of peripheral device nodes in the peripheral device workspace to the smart device node.

10. The peripheral device workspace cloud orchestrator server of claim 9, further comprising: the hardware processor to execute computer-readable program code of a cloud manageability orchestrator module to register and store identification of each of the plurality of device nodes

within the peripheral device workspace and create a manifest of the device nodes to define the peripheral device workspace for association with a peripheral device workspace identification value in a database operatively coupled to the peripheral device workspace cloud orchestrator server.

11. The peripheral device workspace cloud orchestrator server of claim 9 further comprising: the network interface device to receive environmental telemetry data including device node capability data and key performance indicator (KPI) data that includes hardware processing resource utilization at the anchor information handling system node; and the hardware processor executing the computer-readable program code of the telemetry ownership orchestrator module to determine that the hardware processing resource utilization is the resource utilization level that has reached the utilization threshold for a processing resource of the anchor information handling system node.

12. The peripheral device workspace cloud orchestrator server of claim 9 further comprising: the network interface device to receive environmental telemetry data including device node capability data and key performance indicator (KPI) data that includes memory resource utilization at the anchor information handling system node; and the hardware processor executing the computer-readable program code of the telemetry ownership orchestrator module to determine that the memory resource utilization is the resource utilization level that has reached the utilization threshold for a memory of the anchor information handling system node.

13. The peripheral device workspace cloud orchestrator server of claim 9 further comprising: the network interface device to receive environmental telemetry data including device node capability data and key performance indicator (KPI) data that includes network connectivity level resource utilization at the anchor information handling system node; and the hardware processor executing the computer-readable program code of the telemetry ownership orchestrator module to determine that the network connectivity level resource utilization is the resource utilization level that has reached the utilization threshold for a network interface of the anchor information handling system node.

14. The peripheral device workspace cloud orchestrator server of claim 9 further comprising: a cloud orchestrator database to store a peripheral device workspace identification value associated with the peripheral device workspace along with the aggregated KPI data for the peripheral device workspace, a peripheral device workspace manifest, and orchestrated device descriptor (ODD) values that provide features, configurations, and other environmental context data associated with each of the device nodes within the peripheral device workspace.

15. The peripheral device workspace cloud orchestrator server of claim 9 further comprising: the hardware processor to execute computer-readable program code of an ecosystem manageability service module to apply one or more hardware device operational policies to the peripheral device workspace and identify the plurality of peripheral device nodes that form part of the peripheral device workspace and propagate those hardware device operational policies that apply to the plurality of peripheral device nodes.

16. The peripheral device workspace cloud orchestrator server of claim 9 further comprising: the hardware processor to execute computer-readable program code of a workspace context discovery module to monitor for changes in the environmental telemetry data within a peripheral device workspace, update a manifest of the peripheral device workspace, and provide analysis of device node capabilities of each device node within the peripheral device workspace for the workspace context discovery module to map the capabilities; and the hardware processor to execute the computer-readable program code of the telemetry ownership orchestrator module determining, based on the changes in environmental telemetry data and device node capabilities, to alter the instruction to offload the portion of environmental telemetry data collection and aggregation functions to be transitioned from the smart device node within the peripheral device workspace back to the anchor information handling system node when the resource utilization level at the anchor information handling system node is determined to have fallen below a second utilization

threshold.

17. A peripheral device workspace cloud orchestrator server, comprising: a hardware processor; a memory device; a power management unit (PMU) to provide power to the hardware processor and memory device; a network interface device to receive environmental telemetry data including device node capability data and key performance indicator (KPI) data associated with a plurality of device nodes within a peripheral device workspace at a location including an anchor information handling system node; the hardware processor to execute computer-readable program code of a telemetry aggregator module to aggregate the environmental telemetry data associated with the device nodes within the peripheral device workspace; the hardware processor to execute computer-readable program code of a telemetry ownership orchestrator module to map the environmental telemetry data received from the anchor information handling system node to records for each device node in the peripheral device workspace stored in a database; the hardware processor to execute computer-readable program code of a telemetry ownership orchestrator module determine, based on the KPI data and device node capability data, whether telemetry collection and aggregation functions are to be transitioned from the anchor information handling system node to a smart device node within the peripheral device workspace; the hardware processor to execute computer-readable program code of a workspace context discovery module to map the capabilities of each device node within the peripheral device workspace based on a usage context in which a type of peripheral device node among the node devices is needed operate with the anchor information handling system executing an identified software application; the hardware processor to execute computer-readable program code of a workspace capability orchestrator module to, based on the KPI data and usage context associated with the anchor information handling system node executing the identified software application within the peripheral device workspace, provide peripheral device workspace recommendations to select a first device node among a plurality of device nodes having overlapping capabilities within the peripheral device workspace; and the hardware processor to execute computer-readable program code of a cloud orchestrator notification module to generate and transmit to the anchor information handling system within the peripheral device workspace an instruction selecting the first device node and indication that overlapping capabilities associated with other device nodes are no longer available within the peripheral device workspace.

18. The peripheral device workspace cloud orchestrator server of claim 17 further comprising: the hardware processor to execute computer-readable program code of an ecosystem manageability service module to apply the hardware device operational policies to the peripheral device workspace and identify those peripheral devices that form part of the peripheral device workspace having a peripheral device workspace identification value to propagate those hardware device operational policies for application to those peripheral devices.

19. The peripheral device workspace cloud orchestrator server of claim 17 further comprising: a peripheral device workspace cloud orchestrator to create and transmit the hardware device operational policies that define selection among the device nodes with the overlapping capabilities within the peripheral device workspace determined from the received environmental telemetry data for execution of the identified software applications by the anchor information handling system node.

20. The peripheral device workspace cloud orchestrator server of claim 17 further comprising: a cloud orchestrator database to store a peripheral device workspace identification value associated with the peripheral device workspace, the KPI data, a peripheral device workspace manifest, and orchestrated device descriptor (ODD) values that provide features, configurations, and other environmental context data associated with each of the device nodes within the peripheral device workspace.
